

Geo2UBEM: 3D Geometry Abstraction and LLM Multi-Agent Parameter Inference for Automated Urban Building Energy Modeling

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ABSTRACT

Urban building energy modeling (UBEM) is critical for city-scale decarbonization planning. However, most existing calibration workflows, which are deterministic, Bayesian, and machine learning-based, rely on building-level metered energy data as calibration targets. For most urban building stock, such data are unavailable, inconsistent, or inaccessible, excluding most buildings from rigorous, standards-based calibration.

To address the gap, **Geo2UBEM** is introduced as the first end-to-end, meter-free workflow integrating automated scan-to-IDF 3D geometry reconstruction with an LLM-driven multi-agent system (LLM-MAS) for energy parameter inference. Urban 3D geometry is generated from street-level imagery, with orientation-specific window-to-wall ratios refined through facade-aware enhancement. In place of the metered data conventionally required under ASHRAE Guideline 14, CBECS 2018 national median benchmarks are used: the LLM-MAS diagnoses simulation–benchmark discrepancies and iteratively narrows the Bayesian optimization search space until convergence.

Applied to six calibrated campus buildings across four types at Purdue University (101,170 m²), Geo2UBEM achieves ASHRAE-consistent convergence ($|NMBE| \leq 5\%$; $CV(RMSE) \leq 15\%$) *without any metered input*, a result not previously demonstrated for a multi-building cohort. Independent validation against actual metered EUIs confirms all six models fall within CBECS sector ranges, at an average inference cost of \$0.09 per building.

A three-layer sensitivity and uncertainty analysis (Morris screening, Sobol decomposition, Monte Carlo propagation) reveals a cohort-level observability gap: equipment density accounts for 80–99% of annual energy-use variance, while envelope parameters, most directly recoverable from geometry, contribute negligibly. This challenges common geometry-driven UBEM assumptions, indicating calibration effort is better spent characterizing operational loads than refining envelope parameters. Collectively, these results establish a scalable, low-cost pathway to standards-consistent UBEM calibration for urban stocks lacking metered data.

1. Introduction

The building and construction sector represents a significant source of global energy consumption and carbon emissions. According to a 2025 global assessment, this sector accounts for approximately 32% of global energy consumption and 34% of CO₂ emissions [48]. Building operations alone contribute about 30% of final energy consumption and 26% of energy-related emissions worldwide [20]. Reducing emissions from buildings at the city scale requires accurate and detailed models of existing building stocks. Urban building energy modeling (UBEM) addresses this need by simulating the energy demand of individual buildings within campuses, districts, or entire cities [4]. UBEM supports scenario analysis, retrofit planning, and new construction assessment. Additionally, UBEM is increasingly applied in broader urban sustainability initiatives, policy evaluation, and smart-grid applications, including demand response, load forecasting, renewable energy integration, and assessment of energy flexibility [5, 17, 64].

UBEM methodologies can be broadly categorized into two primary paradigms: top-down and bottom-up, with hybrid formulations becoming increasingly important [3, 15]. Top-down models estimate urban energy demand from aggregate socioeconomic and climatic drivers, making them computationally efficient for long-term planning but unable to capture building-level heterogeneity, end-use characteristics, or spatially explicit retrofit impacts. In contrast, bottom-up models explicitly represent individual buildings and aggregate their performance to the district or city scale. Within physics-based bottom-up UBEM, two implementation families are commonly adopted: high-fidelity simulation engines (e.g., EnergyPlus and DOE-2) and reduced-order resistance-capacitance (RC) thermal network models (e.g., TEASER, CitySim, and SimStadt), which trade physical detail for computational efficiency and are widely applied in district heating and urban retrofit planning [59]. While high-fidelity simulations provide interpretable, thermodynamics-based analyses, they require detailed building information and substantial computational resources. Consequently, large-scale UBEM applications typically rely on archetype-based assumptions together with reduced-order models or automated archetype-based model generation to maintain computational tractability, whereas high-fidelity engines are more commonly reserved for representative building analyses,

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model calibration, comparative scenario evaluation, and targeted retrofit studies [15]. In parallel, data-driven approaches based on machine learning have gained increasing attention for their ability to learn building energy-use patterns directly from measured or simulated data. More recently, hybrid approaches have emerged to combine the physical interpretability of simulation models with the predictive efficiency of data-driven methods [28].

1.1. 3D Scan to Building Energy Model

Scan-to-BIM, which refers to the automated reconstruction of Building Information Models (BIM) from point-cloud or image-based 3D scan data, has achieved significant progress in model automation, procedural modeling, and accuracy assessment [43, 2, 6]. In addition, deep learning methods have enhanced large-scale point cloud processing and reconstruction efficiency in construction and geospatial applications [60, 30].

However, advancements in scan-to-BIM do not directly result in simulation-ready building energy modeling. Most scan-based workflows still prioritize geometric fidelity, object recognition, or BIM completeness rather than the subset of parameters required by physics-based energy simulation. Recent studies have started to narrow this gap. O'Donnell et al. [35] demonstrated that LiDAR-derived facade mapping can support building energy performance simulation, while Su et al. [45] showed that unmanned aerial vehicle (UAV) oblique imagery and point-cloud reconstruction can be transformed into high-resolution urban building energy models. Su et al. [44] further demonstrated scalable building reconstruction and window detection for UBEM applications, and Duan et al. [16] showed that exterior BIM for existing buildings can be automated through GIS data and UAV-based 3D modeling. Broader UBEM literature has also highlighted the importance of automated geometric data acquisition for scalable urban simulation [23, 63].

Despite these advances, most existing studies still focus primarily on geometry reconstruction or model generation. They rarely close the full pipeline from urban 3D acquisition to simulation-ready individual-building energy models, and they seldom quantify which parameters remain unobservable from exterior geometry alone. This distinction is critical: the goal is not only to reconstruct buildings, but to identify which simulation inputs can be constrained by urban 3D data and which still require informed inference and iterative refinement.

1.2. Large Language Models in Building Energy Workflows

Although UBEM is not fully automated and mature, at the single-building level, many studies have recently emerged to enable the automation of BEM. Large Language Models (LLMs) have recently emerged to support single-building energy simulation input generation, model interpretation, and related energy analysis tasks, diagnosing model-benchmark mismatch and proposing physically plausible adjustments without relying on metered data. They can be

adapted to complex automated building energy modeling cases through fine-tuning and workflow engineering [22, 37] as candidate tools for automated reasoning in building energy workflows.

A more recent development is the shift from single-prompt interaction toward agentic and multi-agent workflows. Zhang et al. [62] proposed an LLM agentic workflow for automatic building energy model development and debugging, while Lu et al. [31] developed a multi-agent framework for retrofit-oriented automated building energy modeling. Multi-agent LLM frameworks have also been explored for automated energy optimization with unstructured building data [58].

To date, existing studies have primarily focused on document-to-model automation, retrofit analysis, and model debugging. They rarely establish a geometry-first inference workflow for individual-building UBEM, in which urban 3D data serve as the primary source of evidence for model construction and parameter updating. Moreover, existing approaches typically do not translate exterior geometric information (e.g., orientation-dependent envelope cues) into simulation-ready inputs. These limitations motivate to look for new approach for parameter inference.

1.3. Energy Parameter Estimation

Energy parameter estimation for building energy models has been studied extensively at the single-building scale. Foundational reviews categorized optimization approaches into manual, automated, and Bayesian or stochastic families, and showed that most workflows depend on measured building energy data to evaluate model-data mismatch and update uncertain inputs [19, 11, 56]. Recent reviews further emphasize that accuracy is strongly constrained by the type, resolution, and completeness of available observations, rather than by the optimization algorithm alone [18, 12]. This dependency is embedded in both deterministic and probabilistic workflows: deterministic tools such as GenOpt and jEPlus+EA minimize residuals between simulation outputs and measured data [57, 1], while Bayesian inference formulations evaluate the likelihood of candidate parameter sets against observations to infer posterior uncertainty [25, 19, 56].

At the urban scale, this metered-data prerequisite becomes a major bottleneck. Reviews of UBEM applications have consistently shown that large-scale modeling is often limited not by simulation engines, but by the scarcity, inconsistency, and uneven coverage of building-level input and validation data [5, 36, 23]. Bass et al. [8] further demonstrated that large urban-scale building simulations can deviate substantially from measured data when building-specific operational information is unavailable. To address this limitation, prior UBEM studies have relied on proxy strategies, including archetype-based defaults, survey-derived assumptions, or partial utility data. However, these strategies still either require some form of measured information or collapse individual-building variation into population averages [5, 8, 14].

Recent automated UBEM workflows have improved scalable model generation, for example through AutoBEM-style pipelines and geospatially enabled frameworks [34, 63]. Nevertheless, most existing workflows do not jointly meet three requirements central to simulation-ready automated UBEM at the individual-building scale: (1) scalable urban 3D geometry acquisition and simulation-oriented abstraction as the primary input; (2) iterative LLM-assisted diagnosis and refinement of non-geometric energy parameters; and (3) explicit observability analysis to distinguish geometry-constrained parameters from operationally unobservable ones.

1.4. Research Gaps

The preceding review identifies three gaps that existing methods fail to address collectively.

Current UBEM is often inaccurate. Physics-based UBEM without building-specific calibration can deviate substantially from actual energy consumption. Archetype defaults and national survey medians stabilize district-level predictions, but they collapse building-to-building heterogeneity in equipment density, occupancy, and operational schedules [8, 36]. At the individual-building resolution required for retrofit planning and carbon accounting, uncalibrated models are not reliable.

Calibration requires metered data that are rarely available at urban scale. All established calibration approaches — deterministic optimization, Bayesian inference, and machine-learning surrogates — require building-level metered energy records to define a calibration objective [19, 11, 56]. At the single-building scale, end-use sub-metering is often available. At urban scale, most buildings have no such records. Sub-metered electricity, heating, and cooling data remain the exception rather than the norm for city-stock portfolios [5, 8, 64]. The metered-data prerequisite therefore blocks calibration for the majority of buildings that UBEM aims to represent.

Existing calibration and optimization workflows are not designed for urban scale. Single-building calibration is sequential, expert-curated, and computationally intensive; it does not extend to multi-building portfolios without substantial manual effort [34, 5]. No existing workflow jointly provides: (1) automated simulation-oriented geometry abstraction from urban 3D data; (2) LLM-assisted inference of non-geometric, operationally unobservable energy parameters; and (3) probabilistic observability analysis across a multi-building cohort.

1.5. Research Goal and Contribution

To close these gaps, this study introduces Geo2UBEM: a fully automated, end-to-end workflow that converts accessible urban 3D geometry into uncertainty-quantified EnergyPlus models without any building-level metered input. Phase 1 converts multi-source geometry into simulation-ready Input Data Files (IDFs) through a harmonized descriptor pipeline with facade-aware WWR enrichment. Phase 2 deploys a three-agent LLM-MAS system that targets national CBECS benchmarks, enabling accurate and scalable

energy parameter inference for buildings without consumption records.

This study answers the following three research questions:

RQ1. Without building-level metered data across multiple building types, how well can an integrated scan-to-IDF plus LLM-MAS workflow achieve reliable CBECS benchmark convergence-check performance?

RQ2. Under Phase 1 geometry-derived IDFs, how robust is LLM-MAS calibration to initialization choice, and what does cold-start testing reveal about inference robustness and initialization independence?

RQ3. Under Phase 1 geometry-derived inputs, which parameters dominate model uncertainty and observability, and what does this imply for practical energy parameter inference strategy?

The specific research contributions are as follows:

1. **System-level framework contribution.** The proposed Geo2UBEM is an LLM-driven multi-agent system framework that closes the loop from accessible urban 3D geometry to automated EnergyPlus energy parameter inference against national CBECS benchmarks.
2. **Evaluation-framework contribution.** An integrated evaluation framework combines a CBECS benchmark convergence check criterion, to characterize inference robustness, and a three-layer UQ framework (Morris, Sobol, Monte Carlo LHS) to provide statistically grounded observability and uncertainty assessment for further accurate UBEM.
3. **Urban-scale deployment capability.** By decoupling calibration from building-level metered inputs and enabling parallel LLM-MAS inference across buildings, Geo2UBEM exhibits portfolio-scalable performance independent of meter availability.

The remainder of this paper is structured as follows. Section 2 details the Geo2UBEM framework architecture, agent workflow, and UQ protocol. Section 3 presents the case-study configuration and evaluation design. Section 4 reports energy parameter inference performance, UQ results, and cold-start findings. Section 5 interprets the results and discusses broader implications. Section 6 addresses current limitations and directions for future work. Section 7 summarizes the conclusions.

2. Methodology

Figure 1 summarizes the three-component Geo2UBEM framework. Each component is described in the subsections that follow.

2.1. Phase 1: Scan-to-IDF Geometry Pipeline

Phase 1 is a simulation-oriented geometry pipeline rather than full scan-to-BIM reconstruction [43]. Its objective is to produce robust massing descriptors for EnergyPlus, with optional facade-aware enrichments (e.g., orientation-specific WWR and glazing proxies). The pipeline is illustrated in Figure 2.

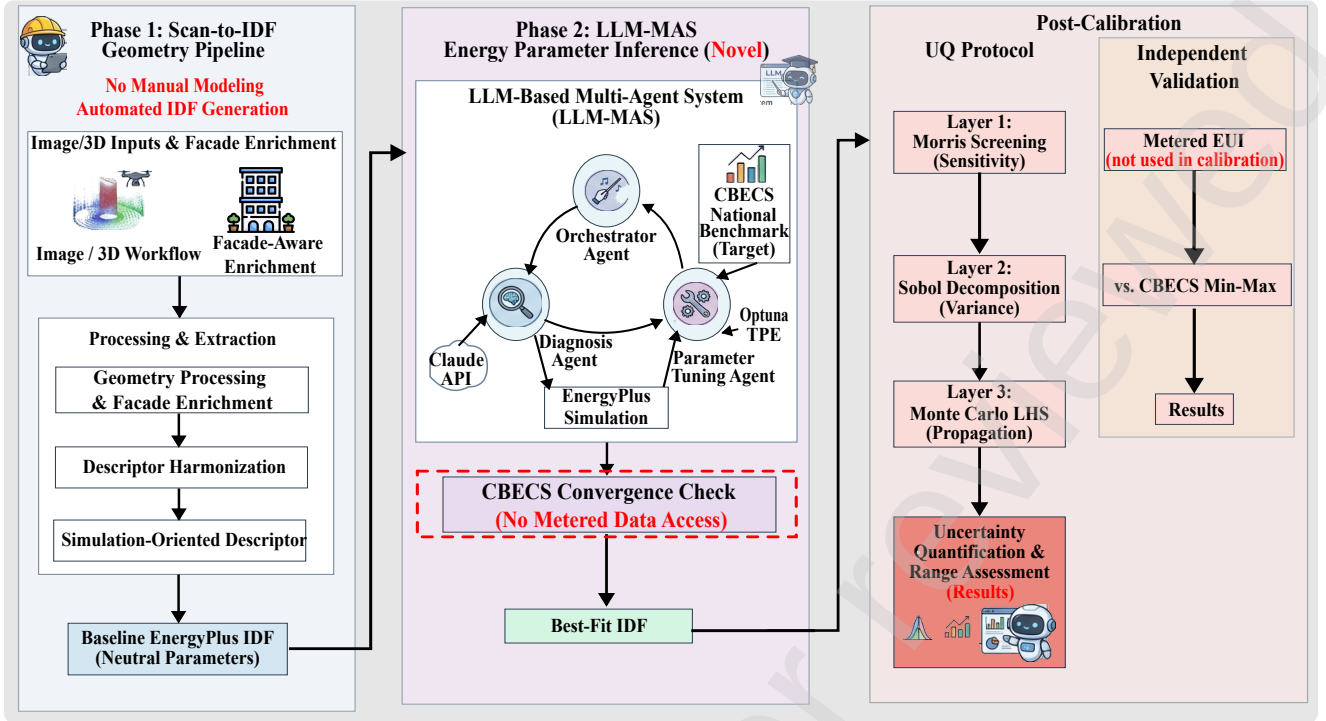


Figure 1: Overall Geo2UBEM framework. Phase 1 extracts simulation-oriented geometry from urban 3D data; Phase 2 infers energy parameters against CBECS 2018 national benchmarks without any metered input; the post-calibration UQ protocol quantifies parameter sensitivity and propagates model uncertainty. The independent validation, performed after calibration is complete, checks whether each building's metered annual EUI falls within the CBECS sector min–max range; metered data are not used at any stage of the calibration pipeline.

Formally, Phase 1 maps source-specific observations G into a simulation-oriented building representation B_s :

$$B_s = f(G) = \{A_f, H, \theta, N_f, A_g, \mathcal{M}\}, \quad (1)$$

where A_f is footprint area, H is building height, θ is principal orientation, N_f is estimated floor count, A_g is gross floor area, and $\mathcal{M}$ is the simplified EnergyPlus-compatible massing model.

When facade-semantic information is available, the representation can be extended to

$$B_s^+ = \{A_f, H, \theta, N_f, A_g, \mathcal{M}, WWR_o, F_o, W_p\}, \quad (2)$$

where WWR_o denotes orientation-specific window-to-wall ratio, F_o denotes facade-level exposure descriptors, and W_p denotes window-placement information. In the present workflow, these fields are optional enrichments rather than mandatory inputs and are obtained through facade-aware processing or reviewed image-derived proxies [45].

2.1.1. Data Acquisition

Phase 1 accepts heterogeneous inputs (image sequences, reconstructed point clouds/meshes, and GIS Facility Metadata). No fixed source is required. In this study, StreamVGGT image sequences were converted to dense 3D geometry [67], and GIS/facility metadata provided priors for scale and orientation checks. These inputs support three purposes: geometry normalization, optional facade enrichment, and

generation of harmonized simulation fields for downstream IDF creation.

2.1.2. Geometry Normalization

Geometry processing and facade enrichment are executed in parallel. The geometry branch normalizes reconstructed shapes into a common local coordinate system; the facade branch converts visual cues into structured envelope proxies.

Step 1 - Image-to-3D reconstruction. Building images are converted into dense 3D point clouds via the learning-based streaming transformer StreamVGGT [67], which are subsequently reconstructed into 3D meshes. This reconstruction preserves roof and facade morphology while capturing overall building massing. Although the geometry predicted by StreamVGGT is initially non-metric (scale is uncertain), it provides sufficient geometric detail for subsequent normalization steps.

Step 2 - Filtering and Cropping. Low-confidence points are removed to reduce noise, and local cropping then isolates the target building. The cropped point cloud is projected onto the horizontal XYplane, and principal component analysis (PCA) is applied to estimate its dominant horizontal directions. Specifically, the eigenvectors of the covariance matrix of the projected coordinates define two orthogonal principal axes, with the eigenvector corresponding to the largest eigenvalue indicating the direction of maximum spatial variance. The point cloud is then rotated

Phase 1: Scalable 3D Geometry Acquisition and Simulation-Oriented Abstraction

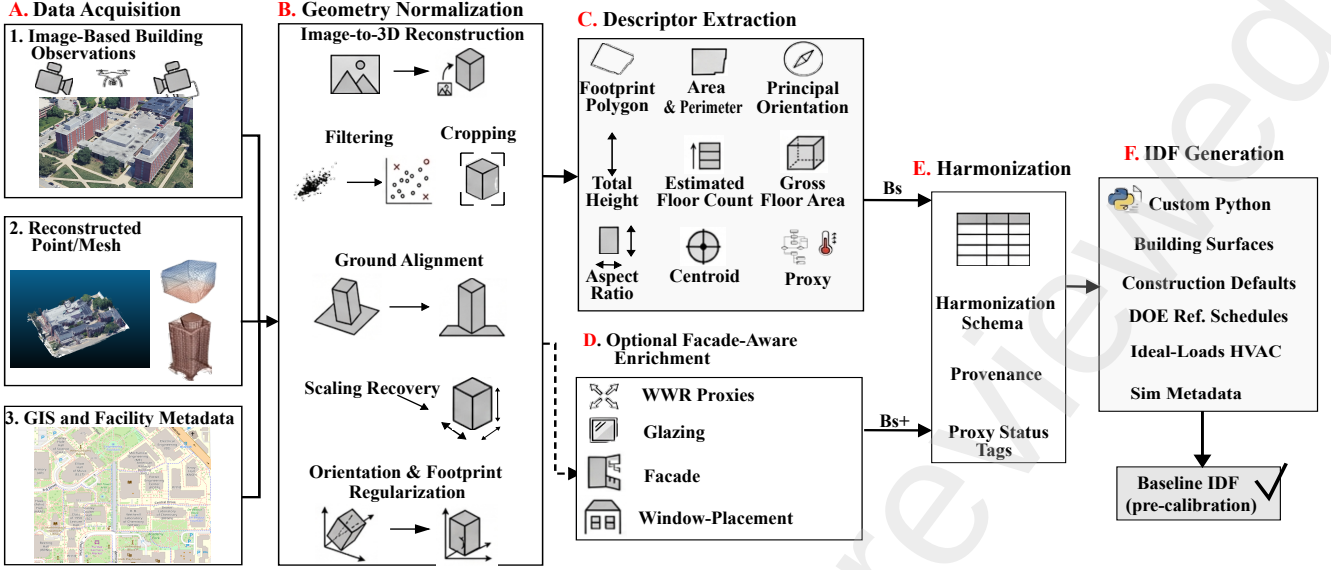


Figure 2: Phase 1 scan-to-IDF pipeline. Panels A–E produce either the geometry-only descriptor B_s or its facade-enriched extension B_s^+ ; Panel D (facade enrichment) is optional, activated only when image data support glazing analysis. Both descriptor variants share the same harmonized schema (Panel E), which decouples upstream acquisition variability from downstream IDF generation and carries provenance metadata (proxy-status tags) for traceability. Panel F converts the descriptor to a neutral baseline IDF $I_0 = g(B_s, \Pi_0)$ using energy-code construction defaults and DOE reference schedules — without any building-specific parameter tuning, so that all parameter fitting is deferred entirely to Phase 2.

about the vertical axis so that the first principal axis aligns with the local x-axis. This coordinate-frame normalization provides a consistent orientation for subsequent footprint extraction and geometric regularization [26, 32].

Step 3 - Ground alignment. A ground plane is estimated by fitting a horizontal plane to the point cloud using RANSAC [42]; a height-based fallback is applied when RANSAC convergence is uncertain. The entire point cloud is then translated so that the detected ground plane aligns with $z = 0$, establishing a consistent vertical reference frame needed for accurate height and floor-count estimation [24, 39].

Step 4 - Scale recovery. Monocular 3D reconstruction is inherently scale-ambiguous. For the scale-invariant outputs of StreamVGGT [67], the absolute scale is resolved by constraining the reconstructed geometry to known building height, footprint area, and aspect ratio from GIS and facility metadata. This yields a metrically accurate 3D representation suitable for energy simulation.

Step 5 - Orientation and footprint regularization. The plan-view footprint is approximated by its minimum rotated rectangle; the long axis of this rectangle defines the principal building orientation, and the short axis defines effective building width. This regularization provides stable, rotationally invariant orientation estimates even for irregular building footprints.

Floor count and gross floor area are then estimated as

$$N_f = \max \left(1, \text{round} \left(\frac{H}{\hat{h}_f} \right) \right), \quad (3)$$

$$A_g \approx A_f \cdot N_f, \quad (4)$$

where $\hat{h}_f$ is a representative storey height derived from facility metadata or building-type archetypes. These estimators are intentionally simple and designed to initialize UBEM massing models rather than to reconstruct detailed floor-level geometry.

2.1.3. Descriptor Extraction

Phase 1 outputs a simulation-oriented descriptor JSON containing footprint polygon, footprint area, perimeter, principal orientation, height, floor count, gross floor area, aspect ratio, centroid, and a coarse thermal-zoning proxy [4, 23]. Harmonization maps upstream geometry and facade outputs into a stable, EnergyPlus-oriented schema with standardized field names, branch provenance, and proxy-status tags [17]; quantitative descriptor accuracy against GIS baselines is reported in Section 4.1. Facade-aware cues are optional and can be replaced by dedicated extraction modules without altering downstream simulation interfaces.

2.1.4. Optional Façade-aware Enrichment

When facade-level information is available, the enrichment module accepts orientation-specific window-to-wall ratio (WWR) values and associated glazing indicators, which are appended to the harmonised descriptor. The module supports two complementary automated approaches: (i) a grammar-based Sobel edge detection pipeline with RANSAC vanishing-point perspective rectification [47], and (ii) semantic segmentation via a fine-tuned SegFormer

(MiT-B5 encoder, IoU = 0.83, mean pixel accuracy = 0.90) applied to georeferenced facade imagery [44].

For the present six-building pilot, building-average WWR was estimated using Method 1 of Tarkhan et al. [47] applied to 11–15 street-level survey photographs per building (63 images in total). The algorithm yielded 22.7%, 21.3%, 22.4%, 24.1%, and 24.0% for MSEE, Dudley Hall, John T. Myers, PMU, and PMU Club respectively (within-building standard deviation $\leq 3\%$). Per-orientation decomposition via SegFormer inference is identified as a direct near-term extension. These fields refine envelope assumptions but are not required for core geometry-to-IDF translation.

2.1.5. Harmonization

Harmonization translates source-specific outputs into a unified schema with standardized field names, provenance metadata, and proxy-status tags. This schema is the stable interface for IDF generation and Phase 2 inference, and supports GIS-based quality checks and downstream observability analysis.

2.1.6. IDF Generation

The harmonized descriptor is translated into a baseline EnergyPlus IDF. Custom Python scripts parse the unified schema and generate building surfaces, optional fenestration placeholders, and baseline simulation metadata. Buildings are represented as simplified thermal massing models using `BuildingSurface:Detailed` objects for floor, roof, and exterior walls. The footprint comes directly from the descriptor, and zone volume is computed from footprint area and assumed storey height. For multi-storey buildings, coarse abstractions (e.g., the Zone multiplier) preserve total floor area and volume without explicitly modeling each storey [10, 11, 50].

Formally, the baseline IDF can be written as

$$I_0 = g(B_s, \Pi_0), \quad (5)$$

where I_0 is the baseline EnergyPlus model, B_s is the harmonized simulation descriptor, and Π_0 is the neutral pre-calibration assumption set. The mapping $g(\cdot)$ converts descriptor fields into EnergyPlus geometry, optional fenestration placeholders, construction defaults, and baseline internal-load/schedule objects.

This step targets simulation readiness rather than geometric completeness. Phase 1 therefore retains only geometry that is most influential for energy simulation, which is sufficient for scalable UBEM massing generation [5].

For the coarse massing model used in Phase 1, the initial building volume is approximated as

$$V_0 \approx A_f \cdot H, \quad (6)$$

where V_0 is initial bulk volume, A_f is footprint area, and H is total building height. This first-order estimate preserves geometric scale for baseline simulation; detailed parameter refinement is deferred to Phase 2.

Construction assemblies follow energy-code defaults for the target climate zone. Occupancy, lighting, and equipment schedules are selected from the DOE reference library by

matched building type [14]. HVAC is represented with EnergyPlus ideal loads (`HVACTemplate:Zone:IdealLoadsAirSystem`) to isolate envelope and internal-load effects from detailed system-model uncertainty [49, 50].

All Phase 1 IDFs are reset to a neutral baseline before calibration. Internal gains and infiltration are assigned conservative reference values, not building-specific tuned values. Detailed baseline settings are reported in Section 3.1. This prevents implicit calibration in Phase 1 and ensures consistent starting points across all calibration phases, with parameter fitting deferred to Phase 2 [11].

2.2. LLM-Driven Multi-Agent Energy Parameter Inference System

Phase 2 replaces meter-driven calibration with benchmark-driven inference against CBECS. An LLM-based multi-agent system iteratively adjusts parameters to reduce benchmark mismatch without using building-level metered data [53, 22, 65, 13]. Figure 3 shows the agent architecture; the convergence check protocol is described in Section 3.4.

2.2.1. Agent Collaboration Workflow

The LLM-MAS comprises three agents (Figure 3): Orchestrator, Diagnosis Agent, and Parameter Tuning Agent. They operate in a fixed loop driven by the Orchestrator.

Orchestrator. The Orchestrator initializes building context (IDF, weather, CBECS target), runs EnergyPlus, evaluates ASHRAE Guideline 14 criteria, and coordinates iterations. If $\text{NMBE} \leq \pm 5\%$ and $\text{CV(RMSE)} \leq 15\%$, the Orchestrator declares convergence and triggers the UQ pipeline. Otherwise, it invokes the Diagnosis Agent for the next iteration.

Diagnosis Agent. The Diagnosis Agent receives the current simulation metrics, end-use energy breakdown, and building context. It calls the Claude API (model : `claude-sonnet-4-6`, temperature = 0.1) with a structured prompt to identify the primary calibration issue and propose a prioritised list of 3–5 parameters. The agent returns a JSON object specifying each parameter's name, adjustment direction, magnitude, and confidence level.

Parameter Tuning Agent. The Parameter Tuning Agent converts the Diagnosis Agent's recommendations into Optuna trial bounds. It then launches a Bayesian Tree-structured Parzen Estimator (TPE) optimisation over 25 trials per iteration, aligning LLM-based diagnosis with automated BES (Building Energy Simulation) search workflows [22, 65, 19]. At each trial, EnergyPlus is called with a sampled parameter set. The trial fitness is a weighted composite:

$$f(\theta) = - (0.3 |\text{NMBE}(\theta)| + 0.7 \text{CV(RMSE)}(\theta)) + 10 \cdot \mathbb{1}_{\text{ASHRAE}}(\theta), \quad (7)$$

where $\mathbb{1}_{\text{ASHRAE}}$ is an indicator that equals 1 when both $|\text{NMBE}| \leq 5\%$ and $\text{CV(RMSE)} \leq 15\%$. The 0.3/0.7 weighting prioritises CV(RMSE) reduction. The step bonus encourages crossing the ASHRAE threshold. Before any LLM-suggested range reaches Optuna, three safeguards are enforced:

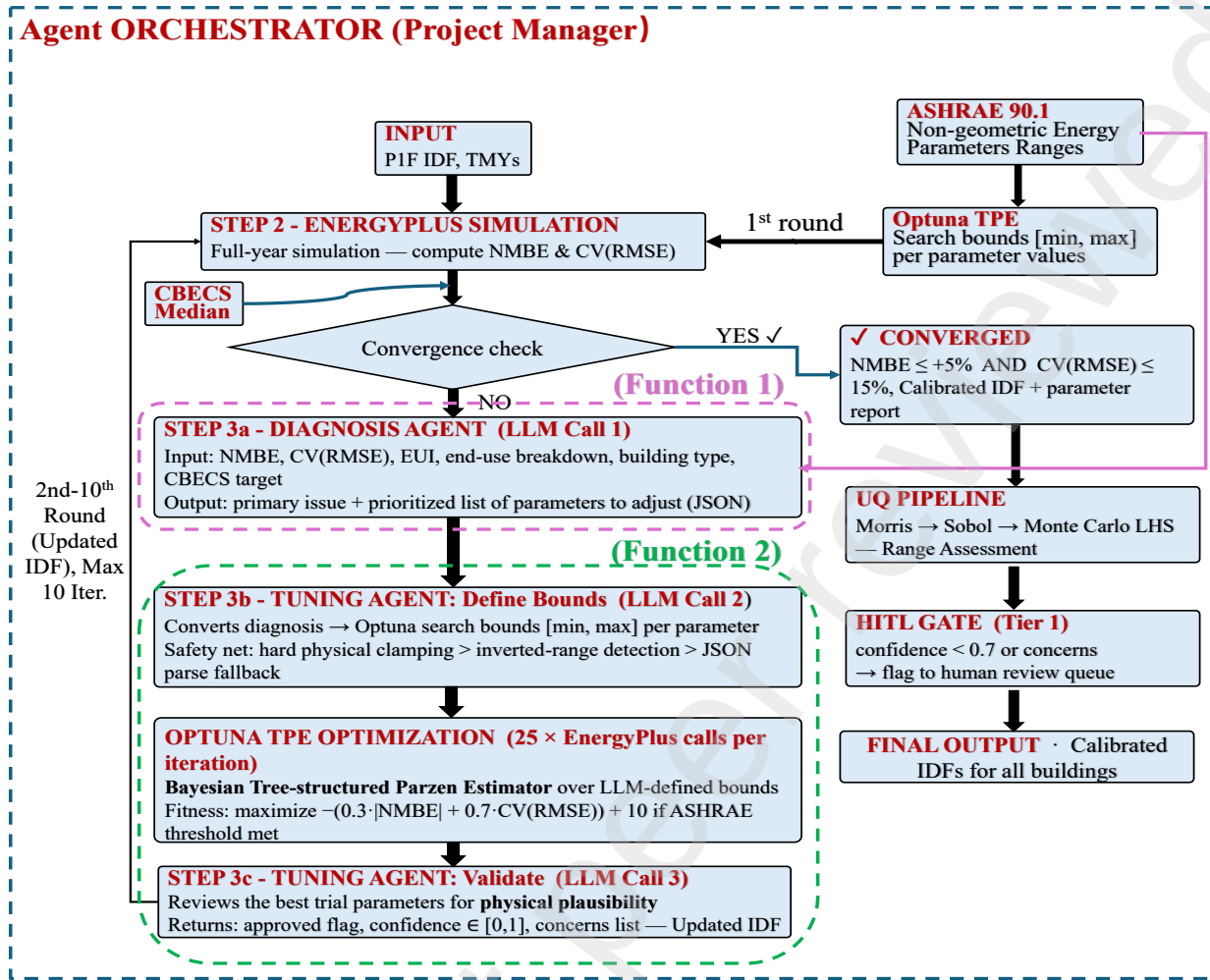


Figure 3: LLM-MAS architecture for Phase 2 energy parameter inference. The two IDF entry paths into the Orchestrator: Iteration 1 supplies the Phase 1 Baseline IDF; Iterations 2–10 supply the Updated IDF produced by the previous cycle. This prevents the loop-back arrow from being misread as overwriting the original input. If the convergence criterion ($\text{NMBE} \leq \pm 5\%$, $\text{CV(RMSE)} \leq 15\%$ vs. CBECS 2018) is not met within 10 iterations, the best-available Updated IDF is passed downstream as a non-converged result. On loop exit, Step 3c confidence scores below 0.7 or a non-empty concerns list route the model to the human-in-the-loop (HITL) gate before entering the UQ pipeline.

1. *Hard physical clamping.* Every suggested $[\min, \max]$ range is truncated to the global physical limits stored in the configuration file (e.g. equipment density $[1, 200]$ W/m²; infiltration $[0.01, 2.0]$ ACH).
2. *Inverted-range detection.* If clamping collapses the range to $\min \geq \max$, the LLM suggestion is discarded and the global configuration bounds are used instead.
3. *JSON parse fallback.* If the LLM response fails to parse, a dedicated `JSONDecodeError` handler, followed by a general `Exception` catch, substitutes the global bounds for any affected parameter.

Empirical fallback frequency and parseability outcomes are reported in Section 4.2.

Structured inter-agent communication contracts. All inter-agent messages follow predefined JSON schemas and are validated before use. The Diagnosis Agent must return `primary_issue`, an ordered `parameters_to_adjust` list, and

optional `additional_considerations`; the Parameter Tuning Agent must return parameter-wise `min/max` bounds with rationale, plus validation fields `approved`, `confidence` $\in [0, 1]$, and `concerns`. This contract-first design reduces parsing errors, enforces bound checking, and improves traceability (Supplementary Table S5).

A two-tier HITL (Human-in-the-loop) safeguard is implemented in the production pipeline. *Tier 1* (*confidence-threshold gate*) is applied immediately after Optuna returns the best-trial parameters. The TuningAgent requests a final LLM plausibility check, which returns an approved flag, a confidence score (0–1), and a concerns list. The Orchestrator routes any building with confidence < 0.7 or a non-empty concerns list to a structured human-review queue. *Tier 2* (*annual spot-check audit*) recommends periodic review of a random sample of passed buildings against stored diagnosis JSON and Optuna trial history. In all cases, the empirically

best Optuna trial is retained; the HITL flag triggers review rather than automatic rollback [31].

Algorithm 1: LLM-MAS Energy Parameter Inference Loop

Input: baseline IDF I_0 , weather file W , CBECS target E^* , $\text{MAX_ITER} = 10$, $n_{\text{trials}} = 25$
Output: calibrated IDF I^* , parameter set θ^* , convergence report

1. Run EnergyPlus(I_0, W) $\rightarrow \hat{E}_0$; compute $\text{NMBE}_0 = (\hat{E}_0 - E^*)/E^*$
2. **if** $|\text{NMBE}_0| \leq 5\%$ **then**
 return (I_0 , CONVERGED at baseline)
3. **for** $k = 1, 2, \dots, \text{MAX_ITER}$ **do**
4. $D_k \leftarrow \text{DiagnosisAgent}(\hat{E}_{k-1}, \text{NMBE}_{k-1}, E^*, \text{building_info})$
 // LLM call 1
5. $B_k \leftarrow \text{TuningAgent.define_bounds}(D_k, \theta_{k-1})$
 // LLM call 2; hard-clamped to physical limits
6. $\theta_k \leftarrow \text{Optuna-TPE}(B_k, n_{\text{trials}} = 25, \text{seed} = 42)$:
 // 25 EnergyPlus simulations
 maximize $[-(0.3 |\text{NMBE}| + 0.7 \text{CV}(\text{RMSE})) + 10 \cdot \mathbb{1}_{\text{ASHRAE}}]$
- over $\theta \in B_k$
7. $v_k \leftarrow \text{TuningAgent.validate}(\theta_k)$
 // LLM call 3: confidence score & concerns
8. Apply θ_k to IDF; Run EnergyPlus $\rightarrow \hat{E}_k$; compute NMBE_k
9. **if** $|\text{NMBE}_k| \leq 5\%$ **then break**
10. **if** $k \geq 3$ and last 2 consecutive CV(RMSE) improvements $< 0.5\%$
 then break
 // early stopping: stuck in local optimum
11. **end for**
12. **if** $v_k.\text{confidence} < 0.7$ or $v_k.\text{concerns} \neq \emptyset$
 then append to HITL review queue
13. **return** ($I^*, \theta^*, \text{report}$)

Orchestrator conflict-resolution policy. The Orchestrator applies four deterministic rules: (i) stateless iteration (no persistent belief state); (ii) approved:false is an audit flag, not a rollback trigger; (iii) corrected_params are logged but not auto-applied; and (iv) directional conflicts are handled by hard-clamping and inverted-range checks before optimization.

2.3. Tool Calling and Closed-Loop Iteration

Data access and information flow. Phase 1 uses building geometry only. Phase 2 has no access to metered data and evaluates output only against CBECS benchmarks.

Phase 2 starts from the raw Phase 1 baseline IDF (neutral defaults). No meter-informed initialization is used. If the baseline already satisfies CBECS alignment at iteration 0, Phase 2 terminates. Otherwise, iterative calibration proceeds as in Algorithm 1.

Parallel execution and tool calling. At each iteration, the Orchestrator runs EnergyPlus, calls the Diagnosis Agent for strategy updates, and triggers Optuna TPE search through the Parameter Tuning Agent. The implementation supports building-level parallelism via worker-level task decomposition. Each worker uses an independent building context and writes to a dedicated per-building Optuna database, avoiding shared-state contention. This architecture supports urban-scale deployment: a portfolio of N buildings can be calibrated concurrently. Under parallel execution, wall-clock time approaches the runtime of the slowest building rather than the sum of individual runtimes. Empirical runtime and speedup results are reported in Section 4.2.3 and Table 5.

2.4. Implementation Details

The Geo2UBEM pipeline was developed in Python 3.11 and evaluated on a 32-core Windows 11 workstation. EnergyPlus 23.2 was the simulation engine. Bayesian optimization used Optuna TPE (seed = 42; 25 trials per pass; n_startup_trials=10; objective: minimize |NMBE|; no pruner). Convergence criteria were $|\text{NMBE}| \leq 5\%$ and $\text{CV}(\text{RMSE}) \leq 15\%$, with $\text{MAX_ITER} = 10$. An early-stop heuristic terminated the loop when the last two consecutive CV(RMSE) improvements were each below 0.5%. Each building used a process-isolated Optuna SQLite database to support concurrent multi-building execution. The LLM layer used Claude Sonnet 4.6 (Anthropic API, claude-sonnet-4-6, temperature = 0.1). Per-building token usage and API costs are reported in Supplementary Table S1.

2.5. Uncertainty Quantification Protocol

Sensitivity analysis (SA) and uncertainty quantification (UQ) are well-established methods for the accuracy analysis of building energy models. Classical methods such as Morris screening and Sobol variance decomposition remain widely used because they balance interpretability and computational cost for high-dimensional parameter spaces [33, 41]. Recent studies continue to apply Morris, Sobol, and uncertainty propagation methods to identify dominant thermal design, occupancy, and operation-related factors, including sensitivity analysis of building thermal design parameters under climate change [46], uncertainty and sensitivity analysis for building energy planning [66], and stock-level uncertainty analysis of building energy models with explicit attention to Sobol convergence and sampling effects [55]. A recent systematic review further confirms that sensitivity analysis is increasingly used to identify influential parameters and reduce model complexity in building energy studies [40]. Yet no prior study has applied global SA to quantify which parameters are structurally unobservable from exterior geometry across a multi-building cohort. We address this gap with the three-layer UQ protocol described below.

After Phase 2 convergence, we applied this protocol (screening, variance attribution, and uncertainty propagation) to all six calibrated models [38, 40]. All UQ hyperparameters were fixed *a priori* using SALib guidance and pilot studies balancing estimator stability and computational cost (Table 1). Figure 4 illustrates the protocol.

2.5.1. Layer 1: Morris Elementary Effects Screening

We used the Morris method [33, 29] to screen parameter importance before Sobol analysis. With r trajectories and p grid levels, Morris requires $r(k+1)$ EnergyPlus simulations per building for k parameters (Table 1). For each parameter i , we computed the mean absolute elementary effect μ_i^* and standard deviation σ_i . Large μ_i^* indicates global influence. Large σ_i relative to μ_i^* indicates nonlinearity or interactions. Sampling and index computation used SALib v1.4 [21].

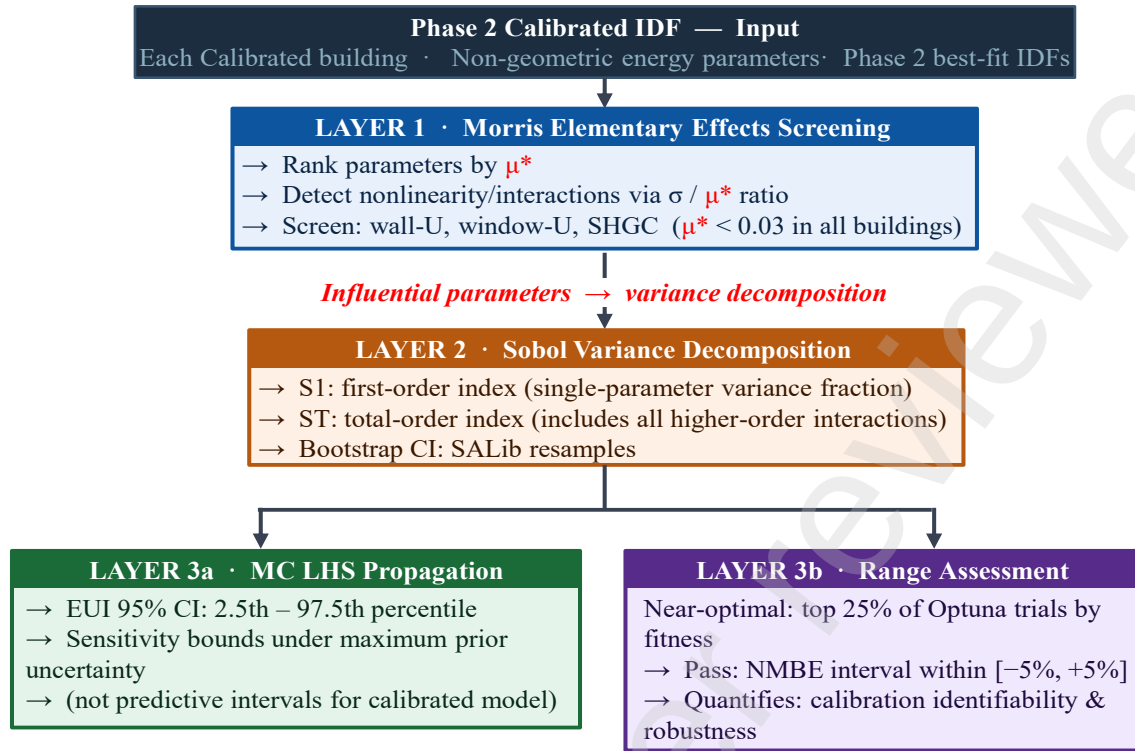


Figure 4: Three-layer uncertainty quantification (UQ) protocol. All hyperparameters (r, p, k, N, n) were fixed *a priori* via SALib guidance and pilot studies (Table 1), not tuned post-hoc. Layer 1 Morris screening serves as a dimensionality-reduction gate: parameters identified as non-influential ($\mu^* < 0.03$ across all buildings) are held fixed in Layers 2 and 3, reducing the active parameter set from $k = 7$ to four. Layer 3b range assessment specifically probes calibration identifiability: a narrow NMBE interval fully within $[\pm 5\%]$ across the near-optimal Optuna subspace indicates a well-constrained solution; a wide or straddling interval signals parameter degeneracy. Numerical results are in Section 4.3 and Figure 7b.

Table 1
UQ protocol hyperparameters ($k = 7$, 6 buildings).

Layer	Symbol	Value	Rationale
Morris (L1)	r (traj.)	10	80 runs/building; sufficient for μ^* ranking.
Morris (L1)	p (levels)	4	Standard for continuous parameters.
Sobol (L2)	N (base)	128	Saltelli: 1 152 runs/building; CI stable at $k = 7$.
MC LHS (L3)	n_{mc}	200	Stable 95% CI; cost scales linearly.
Range Ass.	n_{ra}	20	Lightweight near-optimal compliance probe.

2.5.2. Layer 2: Sobol Variance Decomposition

This layer quantifies variance attribution rather than output-range spread. We computed first-order (S_1) and total-order (S_T) Sobol indices using the Saltelli sampling scheme [41, 55, 61]. With base sample size N and k parameters, the Saltelli $(k + 2)$ -scheme requires $N(k + 2)$ EnergyPlus simulations per building (experimental values in Table 1). S_1 quantifies the fraction of output variance

explained by each parameter alone; S_T includes all higher-order interaction terms. A parameter with $S_T \approx S_1$ acts largely independently. $S_T \gg S_1$ indicates strong interactions with other parameters. Confidence intervals were estimated with SALib's default bootstrap.

2.5.3. Layer 3: Monte Carlo LHS Propagation

This layer estimates output-range uncertainty under parameter uncertainty. We used Monte Carlo sampling with Latin Hypercube design to propagate parameter uncertainty to EUI [66]. We drew n_{mc} samples per building (experimental value in Table 1). When calibrated values are available (after Phase 2 convergence), each parameter is sampled from Triangular($l_i, \hat{\theta}_i, u_i$). Here, l_i and u_i are prior bounds (Table 3), and $\hat{\theta}_i$ is the LLM-MAS inferred value. If calibrated values are unavailable, independent uniform priors over full bounds are used. The 95% confidence interval was computed as the empirical 2.5th–97.5th percentile of the EUI distribution. These intervals represent sensitivity bounds under maximum prior uncertainty; they are not predictive intervals for the calibrated model.

2.5.4. Range Assessment

Range assessment evaluates robustness of the near-optimal solution set rather than global prior-space uncertainty. It quantifies the width of the feasible parameter region satisfying CBECS convergence check thresholds. We identified the near-optimal subspace as the top 25% of Optuna trials by fitness value (lowest |NMBE| vs. CBECS). We then drew n_{ra} Monte Carlo samples uniformly within the near-optimal parameter bounds and computed NMBE for each (experimental value in Table 1). A building passes the range assessment if the full n_{ra} -sample NMBE interval lies within $\pm 5\%$. This protocol indicates whether a solution is tight (narrow interval) or over-parameterized (wide interval spanning the ASHRAE threshold).

2.6. Independent Validation

Because Geo2UBEM calibrates against CBECS national benchmarks rather than building-specific meters, an independent check is required to confirm that the meter-free pipeline produces models consistent with real-world operation. For this purpose, multi-year average metered total site EUI — aggregated from electricity, district chilled water, and district steam records provided by Purdue Facilities Management — is used only after calibration is complete: these records are withheld from the LLM-MAS at every stage of inference and never enter the calibration loop (Figure 1, dashed validation path).

The validation applies two complementary checks. First, each building's metered EUI is positioned within the CBECS 2018 EUI distribution (min, P25, median, P75, max) for its typology; a metered value falling inside the sector min-max range confirms that the CBECS stratum used as the calibration target is a statistically appropriate peer group for that building (results in Table 8). Second, the metered EUI is compared against the Monte Carlo prior envelope from Layer 3 (Section 2.5.3); containment within the envelope confirms that the prior parameter space encompasses the building's actual operating conditions (results in Figure 8).

3. Experimental Design and Case Studies

3.1. Case Description and Data Preparation

Six campus buildings at Purdue University (West Lafayette, Indiana, ASHRAE Climate Zone 5A) as the study cohort were selected. Table 2 lists the building inventory. The cohort spans four occupancy types - education, laboratory, dormitory, and mixed-use food service - and five construction vintages from pre-1946 to 2022, covering a combined gross floor area of 101,170 m². This range was chosen to stress-test the calibration framework across building types with different internal load intensities, HVAC systems, and envelope characteristics [9, 14].

Metered annual electricity, cooling, and heating records were available for all six buildings from Purdue Facilities Management, spanning fiscal years 2020–2025 (three to six years per building, depending on meter availability). For each building, the metered annual EUI used in the independent validation was computed as follows: the three end-use

streams — electricity, district chilled water (cooling), and district steam (heating) — were each converted to site energy in kWh, summed to annual total site energy, and divided by the building's gross floor area (Table 2); the resulting annual EUIs were then averaged across the available metering years to smooth year-to-year operational variability. These records were not accessible to the Phase 2 LLM agent during calibration; they are used exclusively for the independent validation described in Section 2.6. The CBECS 2018 target distributions were matched by occupancy type in Census Division 3. The median was used as the target for the CBECS convergence check. All simulations used the TMY3 weather file for Purdue University Airport (ASHRAE CZ5A, station 724386), consistent with EnergyPlus weather-driven simulation practice [49].

For this study, the Phase 1 IDF baseline used ASHRAE 90.1-2022 Climate Zone 5A construction assemblies: exterior walls (U-0.365 W/m²K, R-20 insulation), roof (U-0.183 W/m²K, R-30 insulation), and double-pane clear glazing (U-1.70 W/m²K, SHGC = 0.25, window-to-wall ratio 30%). All Phase 1 IDFs were reset to a neutral parameter baseline before calibration: equipment density 30 W/m², lighting density 10 W/m², occupancy density 0.05 ppl/m², and infiltration 0.3 ACH. Schedules for occupancy, lighting, and equipment were assigned from the DOE reference building schedule library matched by occupancy type (office, laboratory, dormitory, food service) [14].

Phase 1 data inputs were assembled using a consistent hybrid protocol across the six-building cohort, combining GIS/facility records with image/3D-derived geometry evidence (including StreamVGGT-based reconstruction where available) before harmonization into a common descriptor schema. Quantitative descriptor accuracy results for the full Phase 1 cohort are reported in Section 4.1.

Table 4 reports the zone, surface, and window counts for each Phase 1 facade-aware building. Phase 1 (PIF) IDFs incorporate facade-aware enrichment derived from street-level image analysis: per-orientation window surfaces are added, which accounts for the substantially higher surface and window counts (e.g., DUDL_{P1}: 46 surfaces, 39 windows) relative to the corresponding box models. All models use ZoneHVAC:IdealLoadsAirSystem for HVAC representation.

3.2. Energy Parameter Inference Space

We defined a seven-parameter space for the LLM-MAS (Table 3). Four parameters were modifiable in Geo2UBEM-generated IDFs: equipment density, lighting density, occupancy density, and infiltration ACH. Three envelope parameters (wall U-value, window U-value, SHGC) were defined but not runtime-modifiable. This limitation arises because Geo2UBEM IDFs use detailed WindowMaterial:Glazing and Construction assemblies, rather than the WindowMaterial:SimpleGlazingSystem interface required by the parameter agent. The implications of this interface limitation are evaluated in Section 4.3.

Table 2

Study building inventory and matched CBECS EUI statistics.

ID	Full name	Type	Vintage	Floor area (m ²)	CBECS EUI (kWh/m ²)		
					P25	Median	P75
MSEE	MSEE Building	Education/lab	2014	15200	198.8	279.8	350.8
DUDLEY	Dudley Hall	Laboratory	2022	23713	396.4	589.8	830.5
JOHNT	John T. Myers Res. Hall	Dormitory	1992	19855	282.6	317.2	325.7
PMU	Purdue Memorial Union	Mixed-use [†]	1924/2022	28754	433.4	606.4	828.7
PMUC	PMU Club	Food service	1962	10473	1117.0	1540.0	1998.0
UNIV	University Hall	Education	pre-1946	3175	260.0	335.0	520.0
<i>Total</i>				101170			

[†] PMU composite: food service (30.5%) + public assembly (45.5%) + office (24.0%).**Table 3**

Phase 2 (LLM-MAS) energy parameter inference, prior bounds, and EnergyPlus interfaces.

Parameter	EnergyPlus object	Unit	Prior bounds	Runtime-Modifiable
Equipment density	ElectricEquipment	W/m ²	[10, 120]	✓
Lighting density	Lights	W/m ²	[5, 20]	✓
Occupancy density	People	ppl/m ²	[0.02, 0.25]	✓
Infiltration ACH	ZoneInfiltration:DesignFlowRate	ACH	[0.1, 1.0]	✓
Wall U-value	Material (conductivity)	W/m ² K	[0.3, 2.0]	×
Window U-value	WindowMaterial:SimpleGlazingSystem	W/m ² K	[1.5, 6.0]	×
Window SHGC	WindowMaterial:SimpleGlazingSystem	—	[0.25, 0.75]	×

3.3. CBECS Benchmark Selection

We used the U.S. Energy Information Administration Commercial Buildings Energy Consumption Survey (CBECS 2018) as the Phase 2 benchmark target [52, 53, 54]. For each building, we selected the CBECS principal building activity plus (PBAPLUS) category matching the primary occupancy. We extracted the corresponding EUI distribution (minimum, P25, median, P75, maximum) for the target census division and converted units from kBtu/ft² to kWh/m² ($\times 3.1546$). The calibration objective was $\text{NMBE} \leq \pm 5\%$ against the CBECS median.

For mixed-use buildings, the benchmark target is an area-weighted composite across activity classes:

$$E_{\text{target}} = \sum_{j=1}^m w_j E_j, \quad \sum_{j=1}^m w_j = 1 \quad (8)$$

where E_{target} is the mixed-use composite benchmark EUI (kWh/m²/year), w_j is the floor-area share of activity class j , and E_j is the corresponding CBECS median EUI. Case-specific category selections and weights are provided in Section 3.1; resulting benchmark values are summarized in Figure 5.

3.4. CBECS Benchmark Convergence Check Protocol

The LLM-MAS iteratively adjusts non-geometric energy parameters — equipment power density, lighting density, occupancy density, and infiltration rate — guided by the Optuna Bayesian optimiser (TPE sampler), until the simulated annual EUI converges to the CBECS 2018 sector

median target. Convergence is declared when the following ASHRAE Guideline 14 thresholds are satisfied [7]:

$$\text{NMBE} = \frac{E_{\text{sim}} - E_{\text{ref}}}{E_{\text{ref}}} \times 100\% \leq \pm 5\% \quad (9)$$

$$\text{CV(RMSE)} = \frac{|E_{\text{sim}} - E_{\text{ref}}|}{E_{\text{ref}}} \times 100\% \leq 15\% \quad (10)$$

where E_{sim} is the simulated annual site EUI and E_{ref} is the CBECS sector median EUI. Since each evaluation compares a single simulated annual value against a single reference value, CV(RMSE) reduces algebraically to |NMBE|; the operative criterion is therefore $\text{NMBE} \leq \pm 5\%$, with CV(RMSE) retained for alignment with ASHRAE Guideline 14 notation. No metered data are used at any point during inference or evaluation. A pass confirms that: (1) the LLM-MAS mechanism converges without metered data, and (2) the inferred parameters are physically plausible within nationally representative ranges.

3.5. IDF Configurations and Experimental Variables

The Geo2UBEM pipeline produces Phase 1 facade-aware IDFs (P1F IDFs) as the sole IDF configuration for all Phase 2 experiments. Geometry originates entirely from automated Phase 1 reconstruction (image-driven batch or StreamVGGT polygon extraction). WWR values are assigned by orientation from street-level image analysis, replacing conventional fixed-percentage defaults with per-facade estimates (Section 2.1). All six P1F IDFs share

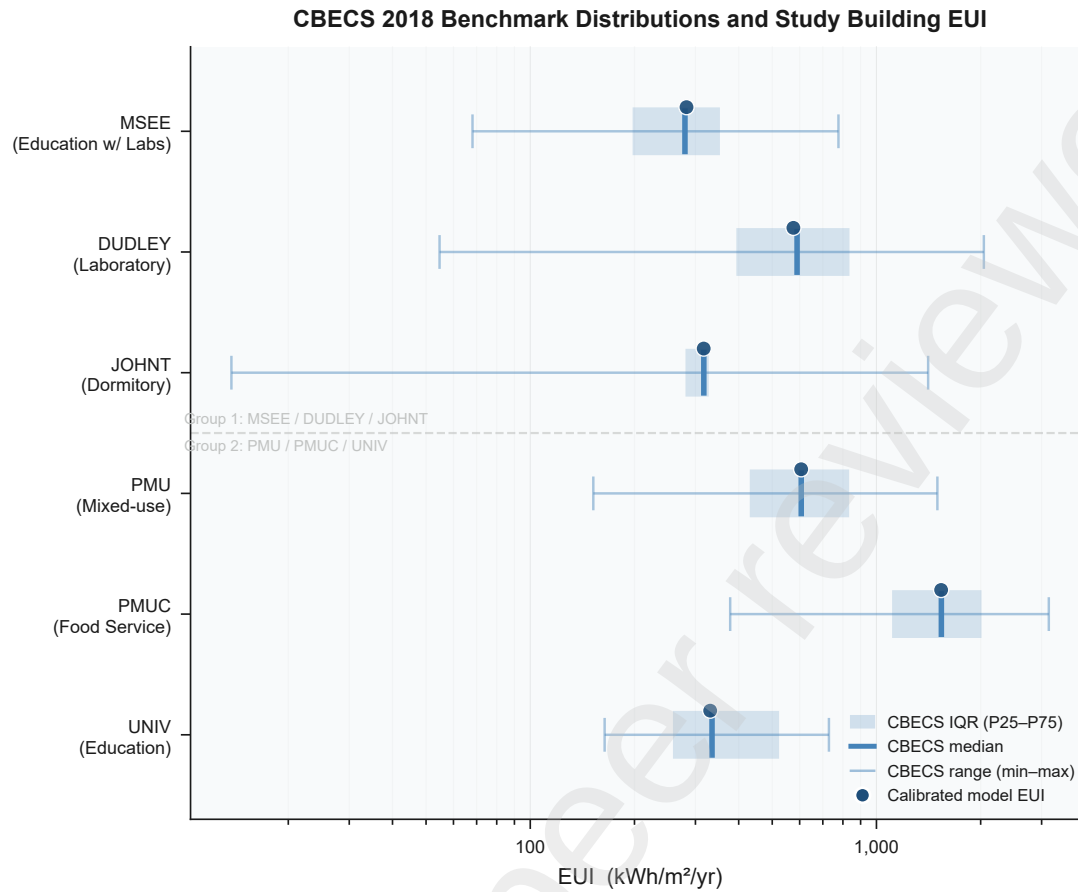


Figure 5: CBECS 2018 EUI distributions (Census Division 3) for six building types. Boxes denote P25–P75, center lines denote medians (benchmark inference targets), and whiskers denote min–max. Mixed-use targets use area-weighted composite medians (Equation 8).

identical neutral load parameters, schedules, HVAC assumptions, and weather file; CBECS sector medians serve as the calibration target (Table 2).

4. Results

We evaluated the proposed Geo2UBEM workflow across six campus buildings at Purdue University, spanning four occupancy types (education, laboratory, dormitory, and mixed-use food service) with a combined gross floor area of 101,170 m². Results are organized into three subsections: (1) Phase 1 geometry pipeline and baseline-IDF assessment; (2) Phase 2 LLM-MAS energy parameter inference performance (Phase 1 facade-aware IDFs, without metered data); and (3) sensitivity analysis and uncertainty quantification revealing parameter observability under geometry-derived inputs. ASHRAE Guideline 14 thresholds ($\text{NMBE} \leq \pm 5\%$, $\text{CV(RMSE)} \leq 15\%$) serve as the acceptance criteria for the CBECS Benchmark Convergence Check.

4.1. Geometry Descriptor and Baseline-IDF Results (Phase 1)

Table 4 summarizes the current Phase 1 outputs for the six Purdue entities. The table reports core geometry metrics and model-structure counts (zones, surfaces, windows) for each building. This comparison should be interpreted as an engineering validation of the Scan-to-IDF pipeline rather than as a claim that all six buildings share the same upstream evidence quality. UNIV remains the most mature image-driven validation case, because it includes both the pure reconstruction route and the GIS-guided refinement comparison. By contrast, PMU and PMUC were processed through a shared-image joint reconstruction followed by footprint-guided building-level separation, and DUDL_LAMB was treated as one metering-aligned combined entity. In Table 4, FP denotes footprint area and GFA denotes gross floor area.

Across the six-building cohort, all recommended Phase 1 baselines were translated into runnable EnergyPlus IDFs. Height agreement is strongest in the cases where reliable override or facility references were available, including MSEE, UNIV, PMU, and PMUC. Floor-count and gross-floor-area alignment were also preserved after override

Table 4

Phase 1 geometry summary and IDF model-structure counts for the six Purdue entities.

Entity	FP (m ²)	FP err. (%)	Height (m)	GFA (m ²)	Zones	Surfaces	Windows
MSEE	3040.00	0.00	14.055	15200.00	1	6	4 [†]
JOHNT	5642.85	-16.78	18.000	28214.24	1	31	26
DUDL_LAMB	3964.02	-7.42	18.000	19820.11	1	46	39
UNIV	899.81	+5.88	15.702	3599.25	1	22	20
PMU	5085.42	-24.47	16.031	23986.64	1	36	30
PMUC	1939.62	-18.40	16.326	10212.37	1	31	20

Nomenclature — FP: plan-view footprint area (horizontal extent from Phase 1 reconstruction); FP err.: percentage deviation of FP from GIS reference polygon; Height: building height from GIS/facility records, used to constrain Phase 1 scale recovery; GFA: gross floor area = FP × estimated floor count; Zones/Surfaces/Windows: EnergyPlus IDF object counts. [†] MSEE's 4 window objects represent orientation-specific equivalent glazing areas (WWR 22–26%); the low count reflects its rectangular footprint, not sparse fenestration.

harmonization for PMU and PMUC, which was important because these two buildings were separated from a shared image set rather than reconstructed from isolated image folders. Footprint agreement remains the main residual source of error, with building-specific causes for each deviation. MSEE achieved zero footprint error (0.00%) because its simple rectangular form was captured cleanly by StreamVGGT reconstruction and confirmed exactly by the Microsoft Building Footprint reference; no boundary ambiguity or wing complexity could introduce residual area. UNIV shows a small positive error (+5.88%), meaning the scan slightly overestimates the footprint; this reflects roof overhangs and covered walkway elements included in the point-cloud boundary during GIS-guided refinement. DUDL_LAMB has a moderate deficit (-7.42%) attributable to its combined treatment as a single metering-aligned entity spanning two physically joined halls, where the shared inter-building boundary introduced minor area underestimation. JOHNT's larger deficit (-16.8%) stems from partial image coverage of its irregular multi-wing footprint, which caused the reconstruction to miss peripheral floor area. PMU (-24.5%) and PMUC (-18.4%) show the largest errors because both buildings were separated from a single overlapping image set rather than reconstructed from isolated folders; footprint boundary ambiguity at the shared wall propagated into systematic area underestimates. These Phase 1 results are sufficient to build runnable baseline IDFs, but geometry improvement alone cannot remove the need for downstream energy parameter inference. More accurate geometry improves simulation readiness and reduces initialization bias, yet it does not guarantee alignment with annual metered end uses, which remain sensitive to internal loads, occupancy schedules, infiltration, and HVAC assumptions — precisely the parameters inferred by Phase 2 LLM-MAS.

4.2. Phase 2 LLM-MAS Energy Parameter Inference

Table 5 summarizes CBECS convergence check results for all six Phase 1 facade-aware IDFs. **Iter.** = 0 indicates the Phase 1 baseline IDF satisfied the convergence check

Table 5

LLM-MAS CBECS Benchmark Convergence Check results: Phase 1 facade-aware IDFs (P1F). All six buildings achieve $\text{NMBE} \leq \pm 5\%$ without metered input.

Building	Type	Area (m ²)	CBECS med. (kWh/m ²)	Conv. NMBE (%)	Iter.	Time (min)
MSEE	Education	15 200	279.8	1.04	7	35.0
Dudley Hall	Laboratory	23 713	589.8	1.00	1	91.5
John T. Myers	Dormitory	19 855	317.2	-0.08	1	63.6
PMU [†]	Mixed-use	28 754	606.4	-0.04	1	70.8
PMU Club	Food svc.	10 473	1540.0	-0.15	3	59.8
University Hall	Education	3175	335.0	-1.22	1	56.3
Convergence check pass rate					6/6	
Wall-clock, parallel 6 workers (min)					91.6	

[†] PMU benchmark is a space-weighted composite: food service (30.5%) + public assembly (45.5%) + office (24.0%). CBECS convergence check criterion: $\text{NMBE} \leq \pm 5\%$ vs. CBECS sector median (no metered data used at any stage). All six measured EUIs fall within the CBECS min-max range (Figure 5).

without LLM-MAS iteration; the Time column reports sequential single-building wall-clock duration.

4.2.1. CBECS Benchmark Convergence Check

All six buildings met the pre-defined CBECS benchmark convergence check thresholds against their respective CBECS 2018 benchmark on both metrics ($\text{NMBE} \leq \pm 5\%$ and $\text{CV(RMSE)} \leq 15\%$) (Table 5, Figure 6; Group 1 in Supplementary Fig. S17). Four buildings (Dudley Hall, John T. Myers, PMU, and University Hall) converged after a single LLM-MAS iteration from the neutral-parameter baseline; PMU Club required three iterations and MSEE seven (Table 5). Final convergence-check NMBE ranged from -1.22% (University Hall) to +1.04% (MSEE); corresponding CV(RMSE) values were identical to |NMBE| within rounding, as expected for annual single-point comparison against a scalar benchmark.

MSEE required seven iterations. Its P1F facade characterization initially pushed the LLM agent toward a high-intensity equipment-density range typical of engineering facilities, whereas its CBECS target (279.8 kWh/m²) is below the education-sector average. This mismatch required iterative downward bound refinement. Each iteration took about 5 min (35.0 min ÷ 7), indicating small, targeted search-space updates rather than broad restarts.

PMU Club exhibited the largest initial bias in the cohort: the neutral baseline underestimated food-service internal loads by -55.1% against the CBECS food-service median (1,540 kWh/m²). The Diagnosis Agent attributed the gap to equipment intensity and raised equipment density stepwise (75.0 → 85.0 → 100.0 W/m², with lighting density reaching 16.0 W/m²), converging at NMBE = -0.15% after three iterations. PMU, calibrated against its space-weighted composite benchmark (food service 30.5%, public assembly 45.5%, office 24.0%; 606.4 kWh/m²), started +15.1% above target and converged to NMBE = -0.04% in a single iteration. Figure 6 shows the Group 2 convergence trajectories (PMU, PMUC, University Hall): NMBE per LLM-MAS iteration (top) and the corresponding simulated EUI approaching the

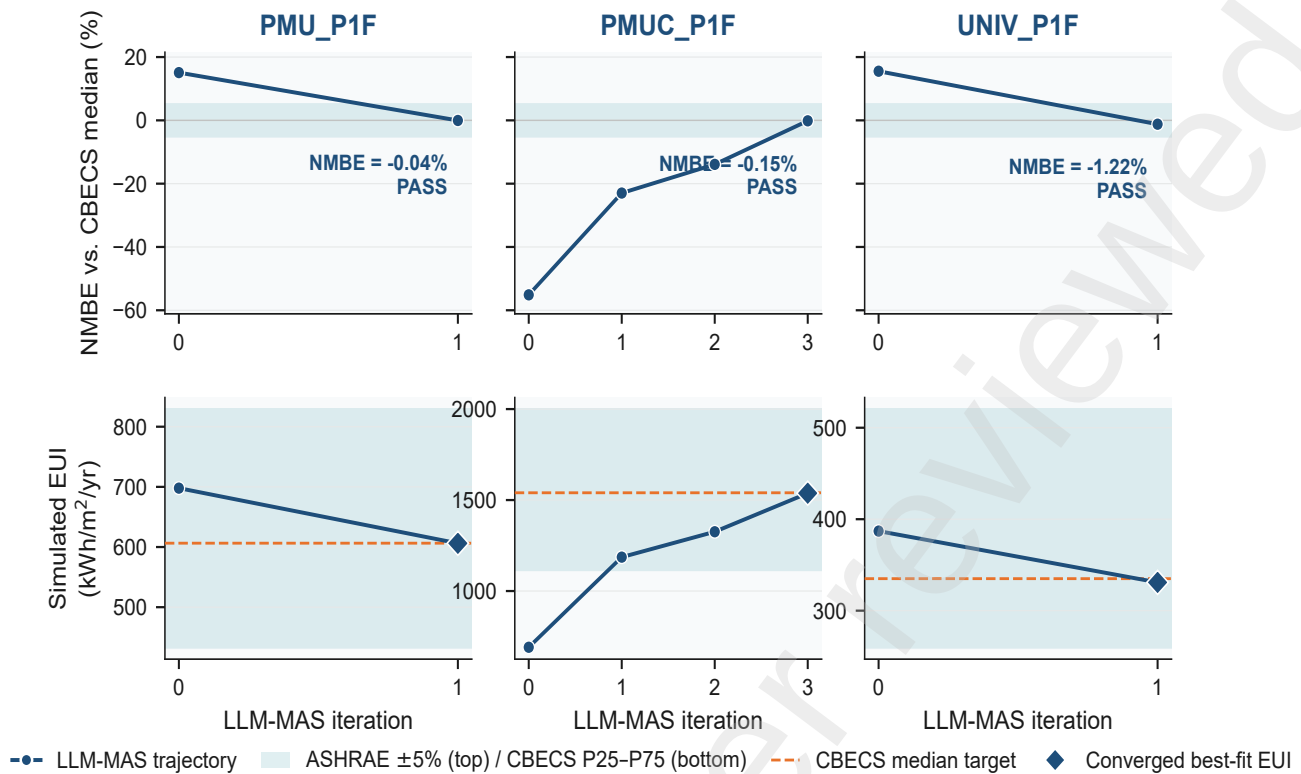


Figure 6: LLM-MAS convergence for P1F Group 2 (PMU_P1F, PMUC_P1F, and UNIV_P1F). Top: NMBE versus CBECS sector median at each LLM-MAS iteration; the shaded band marks the ASHRAE $\pm 5\%$ acceptance range and iteration 0 is the neutral-parameter baseline. Bottom: the corresponding simulated EUI trajectory converging onto the CBECS median target (orange dashed line); the shaded band marks the CBECS P25–P75 range and the diamond marks the converged best-fit EUI. All three buildings reach $|NMBE| \leq 5\%$ within three iterations without metered data, including PMUC, whose initial -55% bias — the neutral baseline severely underestimating food-service internal loads — was closed through three successive equipment-load adjustments.

CBECS median target (bottom); Group 1 (MSEE, Dudley Hall, John T. Myers) is shown in Supplementary Fig. S17 with identical layout.

4.2.2. Calibrated Energy Parameter Values

Table 6 reports the finalized energy parameter values inferred by LLM-MAS for each of the six P1F buildings. Four parameters were modifiable at runtime: equipment density, lighting density, occupancy density, and infiltration ACH. The Diagnosis Agent adjusts a parameter only when its diagnosis ranks that parameter as necessary for closing the benchmark gap; parameters not flagged remain at the neutral-baseline value defined in Section 3.1, shown in italics in the table. The three envelope parameters were not runtime-modifiable in the current IDF interface and were fixed at the ASHRAE 90.1-2022 CZ5A construction values given in Section 3.1 for all six buildings (Section 3.2); values are restated in the table caption for reference.

4.2.3. Calibration Efficiency and Scalability

The P1F pipeline was benchmarked on a single 32-core workstation in parallel mode (one worker per building) to assess scalability toward urban-scale deployment (Table 5). Wall-clock time was **91.6 min** for six buildings, with Dudley

Table 6

Finalized LLM-MAS inferred values of the runtime-modifiable energy parameters for the six Phase 1 facade-aware IDF. Italicised entries denote parameters the Diagnosis Agent left at the neutral-baseline value (no adjustment required for convergence). Envelope parameters were not runtime-modifiable and were fixed at ASHRAE 90.1-2022 CZ5A values for all buildings (wall U-0.365, roof U-0.183, window U-1.70 W/m²K; SHGC = 0.25).

Building	Equip. density (W/m²)	Lighting density (W/m²)	Occupancy (ppl/m²)	Infiltration (ACH)
MSEE	10.0	5.1	<i>0.05</i>	<i>0.30</i>
Dudley Hall	24.0	10.1	<i>0.05</i>	0.299
John T. Myers	10.1	8.3	0.093	0.349
PMU	29.6	10.6	<i>0.05</i>	0.225
PMU Club	100.0	16.0	<i>0.05</i>	0.317
University Hall	119.9	15.2	<i>0.05</i>	0.206
Prior bounds	[10, 120]	[5, 20]	[0.02, 0.25]	[0.1, 1.0]

Hall (DUDL_P1F, 91.5 min) as the bottleneck owing to its large floor area (19 820 m²). MSEE_P1F required 7 calibration iterations yet completed in only 35.0 min, confirming that iteration count alone does not determine wall-clock

time. The estimated sequential total is ≈ 377 min ($\approx 4.1 \times$ parallel speedup).

For contextual interpretation, prior studies report that expert manual model authoring can require hours per case [22]. However, we did not perform a controlled manual-baseline experiment (e.g., IDF Editor/OpenStudio) on the same six cases; therefore, literature-reported manual times are used only as background context and not as a directly comparable benchmark for quantitative speedup claims.

These results confirm that the pipeline scales linearly with available hardware: wall-clock time converges to the single-building maximum (~ 92 min for P1F IDFs) when one process is allocated per building. The ceiling reflects per-building simulation cost driven by facade geometry complexity, not iteration count.

The LLM reasoning layer incurs a modest and predictable overhead: **\$0.09 USD per building** on average (Supplementary Table S1), totalling under \$0.55 for the six-building campaign — negligible relative to the ~ 92 -min parallel wall-clock cost. The system required no manual parameter tuning, no metered data, and no building-specific configuration beyond the IDF and CBECS category.

4.3. Sensitivity Analysis and Uncertainty Quantification

4.3.1. Parameter Sensitivity: Morris Screening and Sobol Indices

Sobol total-order indices (S_T) consistently identified equipment density as the dominant driver of EUI variance across all building types (Supplementary Table S3, Figure 7b). S_T for equipment density ranged from 0.797 (Dudley Hall) to 0.994 (University Hall), meaning 80–99% of total output variance was attributable to this single parameter. Lighting density ranked second in five of six buildings ($S_T = 0.011$ – 0.013); occupancy density ranked second in Dudley Hall ($S_T = 0.136$). This elevated occupancy sensitivity is consistent with Dudley Hall's laboratory classification: laboratory buildings have high people-driven ventilation loads, so occupant density directly affects cooling and fresh-air conditioning energy beyond the plug-load effect captured by equipment density alone. Infiltration ACH contributed negligibly in five buildings ($S_T < 0.001$) but reached $S_T = 0.033$ in Dudley Hall, consistent with its high ventilation-driven equipment intensity. Envelope parameters (wall U-value, window U-value, SHGC) were effectively non-contributing ($S_T < 0.005$) in all buildings, confirming that envelope parameterisation is of low priority for this internally-load-dominated building stock. S_1 for equipment density is nearly identical to its S_T value ($S_1 \geq 0.97 S_T$), confirming an independent main effect with negligible higher-order interaction. Supplementary Table S3 reports full per-building indices.

4.3.2. Monte Carlo Uncertainty Propagation

Monte Carlo sampling with Latin Hypercube Sampling (LHS) over the full prior parameter bounds (200 simulations per building) produced 95% confidence intervals covering

the central range of the CBECS distribution for each building type, with lower bounds generally near the CBECS P10 and upper bounds near or above the P90 (Figure 8a). The violin body represents the probability density of the simulated EUI values, while the shaded band and dotted lines denote the P2.5–P97.5 interval. The dashed line indicates the sample mean, the filled navy diamond (◆) marks the calibrated best-fit EUI, the orange star (★) denotes the multi-year average metered site EUI from Purdue Facilities Management (used only for independent validation, after calibration), and the dash-dot line shows the CBECS median target. Each panel corresponds to one study building, with the horizontal axis serving only to separate simulation samples. The measured EUI falls within the Monte Carlo prior envelope for all six buildings, indicating that the prior parameter space encompasses plausible real-world operating conditions. For MSEE, PMU, and PMUC, the measured EUI differs noticeably from the calibrated EUI, reflecting building-specific departures from the CBECS sector median, as discussed in Section 4.4.2. These envelopes represent sensitivity bounds under broad prior uncertainty rather than prediction intervals of the calibrated model.

4.3.3. Range Assessment

Whereas the Monte Carlo analysis in Section 4.3.2 samples the full prior parameter space to quantify total prior uncertainty, the range assessment evaluates robustness within the already-identified near-optimal region. Specifically, 20 parameter sets were drawn by LHS from the top-25% of Optuna TPE trials ranked by fitness (lowest NMBE against the CBECS median), and each was re-simulated to compute NMBE. This tests whether benchmark compliance holds across the neighbourhood of good solutions, not just at the single best-fit point. All six P1F buildings produced NMBE intervals fully within the ASHRAE $\pm 5\%$ band, with spreads spanning < 6 pp (Figure 8), demonstrating that near-optimal Optuna parameter combinations consistently satisfy benchmark compliance across all six building types.

4.4. Framework Robustness

4.4.1. Cold-Start Experiment

To verify that Phase 2 LLM-MAS inference does not depend on a warm-start IDF initialization, all six buildings were re-calibrated from neutral ASHRAE 90.1-2022 CZ5A code-minimum defaults.

CBECS convergence check results. All six buildings met the CBECS benchmark convergence check thresholds from the neutral starting point (Table 7). Five buildings converged within 1–2 LLM-MAS iterations; PMUC required 5 iterations due to the high equipment intensity of the food-service category. University Hall passed the convergence check on the Optuna baseline without requiring any LLM-MAS iteration (NMBE = -3.89%), indicating that the neutral ASHRAE default parameter set already fell within the CBECS benchmark range for this building type. This zero-iteration convergence reflects an inherent consistency

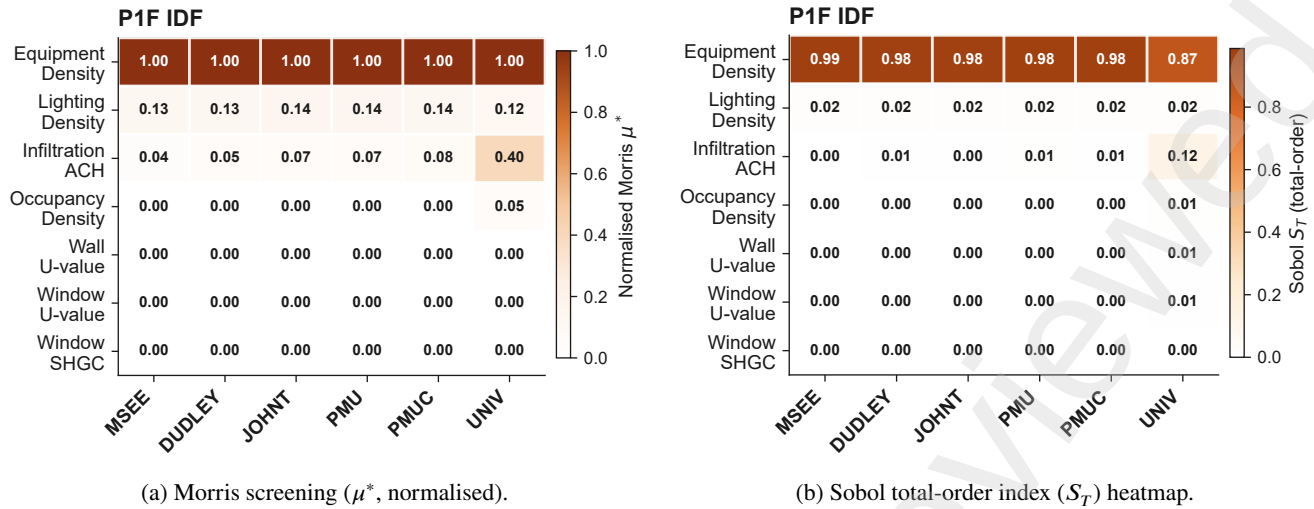


Figure 7: Global sensitivity diagnostics for Phase 1 facade-aware IDFs. Panel (a) shows Morris elementary effects screening (μ^* , normalised), and panel (b) shows Sobol total-order indices (S_T) across the six buildings.

Table 7

Cold-start (neutral ASHRAE defaults) versus standard Phase 1 IDF-initialized convergence-check results.

Building	Standard (P1F initialized)		Neutral (cold start)	
	Conv.	NMBE	Conv.	NMBE
MSEE	+1.04%	7	-0.23%	2
Dudley Hall	+1.00%	1	-0.15%	2
John T. Myers	-0.08%	1	+0.02%	1
PMU	-0.04%	1	-0.20%	1
PMUC	-0.15%	3	-0.02%	5
Univ. Hall	-1.22%	1	-3.89%	0
Pass rate	6/6		6/6	

between ASHRAE 90.1-2022 CZ5A code-minimum defaults and the CBECS education median for small temperature-climate buildings: the code-baseline internal-load assumptions (equipment density 30 W/m², lighting 10 W/m²) produce EUI estimates that fall near the CBECS sector median by design, since both benchmarks draw on overlapping building-stock surveys.

Comparison with standard-initialized runs. Table 7 compares cold-start results against standard Phase 1-initialized runs (Section 4.2). NMBE difference between initializations is 1.2 pp. The largest deviation (4.5 pp, University Hall) arose because the cold-start Optuna search found a passing solution without LLM-guided bound refinement, landing on a different but equally valid parameter set in the flat objective landscape near the CBECS education median. These results confirm that LLM-MAS inference is initialization-independent: the CBECS benchmark target drives convergence regardless of the starting parameter set.

Table 8

Independent validation: metered annual total site EUI of each study building positioned within the CBECS 2018 EUI distribution for its building typology. Metered values are multi-year averages from Purdue Facilities Management records and were not used in calibration.

Building	Type	Metered EUI	CBECS 2018 EUI (kWh/m ²)			Position in distribution
			P25	Median	P75	
MSEE	Educ.+Lab	529	199	280	351	P75-max
Dudley Hall	Laboratory	607	396	590	831	P25-P75
John T. Myers	Dormitory	295	283	317	326	P25-P75
PMU	Mixed-use	1,071	433	606	829	P75-max
PMU Club	Food Svc.	1,092	1,117	1,540	1,998	min-P25
Univ. Hall	Education	357	260	335	520	P25-P75
Within CBECS range		6/6	within min-max;		3/6	within P25-P75

Metered = multi-year average total site energy (electricity + district chilled water + district steam). CBECS source: 2018 CBECS Table C12 (Census Division 3); typology per PBPLUS code.

4.4.2. Metered EUI vs. CBECS Benchmark Distribution

To assess whether the CBECS 2018 national EUI distribution is a statistically appropriate benchmark for this campus cohort, Table 8 compares the multi-year average metered total site EUI of each study building against the CBECS P25, median, and P75 percentiles for its respective typology. Metered values were obtained from Purdue Facilities Management records (electricity, district chilled water, and district steam) and were retained solely for this independent validation; they were not used at any stage of the Geo2UBEM calibration pipeline (validation protocol in Section 2.6).

All six study buildings fall within the observed CBECS min-max range for their respective typologies, confirming that the CBECS 2018 distribution statistically captures the energy behaviour of this campus cohort. Three buildings

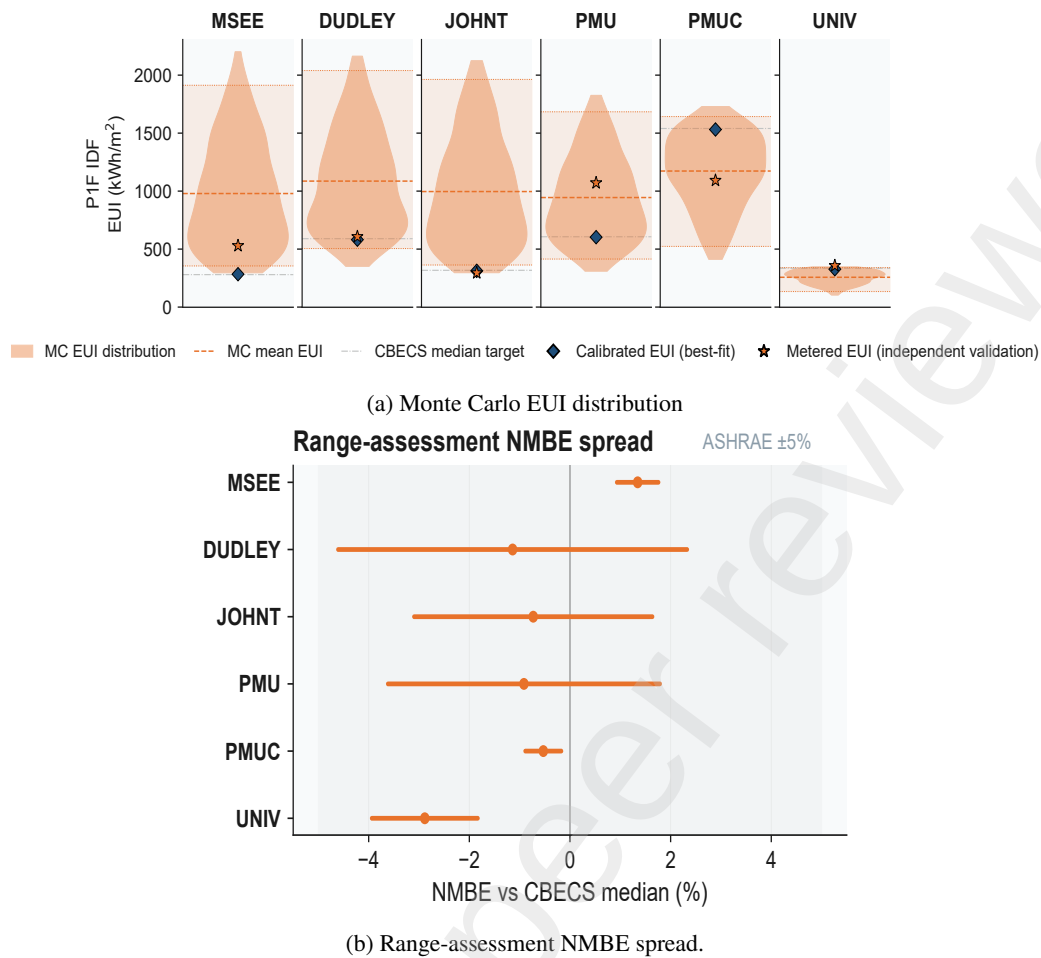


Figure 8: Uncertainty propagation and robustness diagnostics for Phase 1 facade-aware IDFs. Panel (a) shows Monte Carlo EUI distributions under full prior parameter uncertainty; panel (b) shows NMBE spread within near-optimal Optuna bounds.

(Dudley Hall, John T. Myers, and University Hall) lie within the P25–P75 interquartile range; MSEE and Purdue Memorial Union sit above P75, reflecting high equipment-density laboratory and mixed-use operational intensity; PMU Club falls below P25 in the food-service stratum, consistent with its smaller conditioned footprint relative to full-service restaurant benchmarks. The median therefore represents the minimum-bias calibration target for this heterogeneous cohort: it is the centre of a distribution that encompasses all six buildings, and neither the P25 nor P75 extreme would serve as a lower-bias population reference across the four occupancy types studied [52].

5. Discussion

This section discusses the broader implications of the Geo2UBEM framework. We interpret the shift from meter-dependent parameter optimization to benchmark-driven LLM-MAS energy parameter inference, analyze the physical observability gap inherent in geometry-derived UBEM inputs, evaluate the role and constraints of LLM reasoning in the inference loop, and synthesize current limitations with corresponding future research directions.

5.1. LLM-MAS Energy Parameter Inference as a Paradigm Shift for UBEM

The key finding of this study is an observability gap. Sobol decomposition across six buildings shows that parameters geometry cannot supply (equipment density and occupancy) dominate EUI variance, whereas geometry-derived parameters (envelope U-values and SHGC) contribute little. This reflects a structural feature of internally-load-dominated buildings in temperate climates, not a limitation of any specific algorithm in the pipeline. Geo2UBEM made this finding visible. In a controlled, benchmark-driven experiment across six building types, a single building can show the gap, but only a multi-building design reveals it as a stable pattern.

Traditional energy parameters are identified by using meter data and optimization: the optimizer minimizes the gap between simulation and measurement [19], and the energy models are tuned to match each building’s consumption history. At urban scale, this prerequisite is rarely met: annual end-use metered data are unavailable for most buildings, and heating and cooling submeter data are even scarcer.

LLM-MAS removes the metered-data prerequisite by inferring energy parameters against the CBECS 2018 national benchmark. The CBECS convergence check shows that the mechanism operates as intended: all six P1F buildings reached benchmark convergence ($\text{NMBE} \leq \pm 5\%$, $\text{CV(RMSE)} \leq 15\%$) without any metered input. This is a convergence result, not a direct validation against building-level measurements, because both optimization and evaluation use CBECS targets. Even so, the results indicate that LLM-guided Bayesian search can reliably identify physically plausible parameter sets that reproduce nationally representative energy-use levels, including from neutral cold starts (mean $|\Delta\text{NMBE}| = 1.2$ pp; Section 4.4.1). In practice, LLM-MAS places each model within the statistical distribution of comparable national peers, which is an appropriate objective for urban energy planning where fleet-level EUI characterization matters more than building-specific fault detection. Achieving building-specific accuracy will require either building-level meter data or additional operational parameters that separate above-median from below-median operating intensity.

5.2. The Observability Gap: Geometry vs. Operational Parameters

Point cloud scanning and BIM-derived geometry provide reliable inputs for a subset of UBEM parameters: zone footprint, storey height, floor count, and building orientation. These quantities govern envelope heat transfer and determine simulation geometry with little ambiguity. However, Sobol variance decomposition reveals a strong asymmetry in this cohort. Equipment density — the parameter least observable from exterior geometry — explains 80–99% of total EUI variance across all six buildings ($S_T = 0.797$ – 0.994). In contrast, the envelope parameters most directly informed by Phase 1 geometry (wall U-value, window U-value, SHGC) contribute negligibly ($S_T \leq 0.005$ in all cases).

This asymmetry defines an *observability gap*: exterior scans capture the geometry that matters least for energy prediction, while the parameters that matter most — occupant density, plug loads, and operational schedules — are invisible to any remote sensing modality. Accurate zone geometry remains a prerequisite for physically valid simulation, but geometry alone cannot close the calibration problem. Closing the gap requires either metered data or a statistically grounded surrogate such as CBECS benchmarks (Phase 2's approach). CBECS benchmarks serve as this surrogate: they embed the aggregate effect of typical occupant behavior and equipment load intensity for each building type, providing an informative prior without requiring building-level measurement.

The observability gap has a direct implication for scan-to-model pipeline design: improving envelope reconstruction alone is unlikely to deliver the largest gain in EUI accuracy for this cohort. More effort is commonly directed

at window detection, wall thickness estimation, and glazing-ratio inference. While valuable for other applications (architectural visualization, structural assessment), the evidence from this cohort suggests that comparable effort invested in estimating internal load density — from utility billing records, floor use surveys, or satellite-inferred occupancy signals — would yield larger reductions in EUI prediction error.

The observability gap finding is, however, contingent on the building cohort, model structure, and climate studied here. All six buildings are in ASHRAE Climate Zone 5A and span education, laboratory, dormitory, and food service — types whose energy use is dominated by internal loads rather than envelope-driven heating and cooling. In climates where envelope transmission dominates (e.g., extreme cold or hot-dry), or for buildings undergoing envelope retrofit, envelope parameters may carry higher Sobol indices and the observability gap would narrow correspondingly. Generalization to other climate zones or typologies requires additional validation.

5.3. The Role and Constraints of LLM Reasoning in the Calibration Loop

The PMU case isolates the practical value of LLM reasoning beyond parameter optimization (Section 4.2). The initial simulation missed the CBECS benchmark not because of incorrect parameter values, but because a food-service-only benchmark was structurally inappropriate for a mixed-use building. A conventional optimizer minimizes a user-defined objective but does not question whether the objective itself is appropriate. Revising it — from a single archetype target to a mixed-use composite — requires analyst intervention. The Diagnosis Agent performed this meta-level revision autonomously: it reasoned about building function, cross-referenced CBECS categories, and reformulated the target as a space-weighted composite, enabling benchmark-aligned convergence without metered input.

At small scale, experts can make this correction manually. At urban scale, repeating the same diagnostic step across many buildings is labor-intensive. LLM-assisted diagnosis therefore offers a practical scalability advantage.

The same logic extends beyond PMU. It can handle other objective-definition errors, such as mixed-use buildings, facilities with atypical program intensity, or buildings whose use class has changed over time.

This benefit comes with transparency constraints. The Claude API (claude-sonnet-4-6) is a black-box inference step: it returns structured JSON outputs, but its intermediate reasoning is not formally verifiable. In this study, the PMU reclassification was validated post hoc against Purdue University records. Appendix A.1 provides a representative PMU interaction, including the Diagnosis Agent output and the resulting LLM-defined search space. For deployment, reasoning logs and sensitivity flags should be expanded beyond the current three-layer HITL confidence gate (Section 2.2) to also audit objective-reformulation decisions.

LLM-guided diagnosis also addresses a failure mode distinct from parameter mis-specification. Near-zero total NMBE can coexist with large end-use residuals when aggregate-objective optimization masks errors through cancellation. The multi-agent loop mitigates this by reasoning about objective suitability and building-type-specific bounds, not only by improving numerical search.

The computational overhead of this reasoning layer is modest. Worker-level isolation with per-building Optuna databases avoids shared-state contention. In the six-building run, parallel execution achieved a $4.1\times$ wall-clock speedup (377.0 min serial to 91.6 min parallel; Table 5). Practical bottlenecks were dominated by EnergyPlus runtime for multi-zone models and by upstream IDF preparation, rather than LLM orchestration.

6. Limitations and Future Work

Several limitations bound the generalisability of these findings; each points to a concrete research extension.

1. **Cohort scale and climate coverage are limited.** Six buildings in a single climate zone (CZ5A) constrain statistical representativeness. Future work will scale to 15–20 buildings across multiple occupancy types, construction vintages, and at least two climate zones. This expansion will enable formal tests of whether post-calibration residuals correlate with each building's deviation from its CBECS sector peers.
2. **HVAC realism remains simplified.** Ideal-loads systems reduce modelling uncertainty but suppress equipment-level dynamics and contribute to simulation-adequacy gaps in district-heating-dominated buildings. Future implementations will integrate equipment-specific HVAC templates selected through metadata- and agent-assisted classification, improving operational thermodynamic realism without sacrificing workflow automation.
3. **Envelope parameter mutability is restricted in the current IDF interface.** For this internal-load-dominated cohort, Sobol indices confirm that envelope parameters have negligible EUI sensitivity ($S_T \leq 0.005$). However, envelope calibration may be decisive in other climates or typologies. Future work will extend the parameter interface to support runtime calibration of wall and window thermal properties — including simplified glazing objects — enabled automatically when sensitivity screening identifies envelope dominance.
4. **LLM reasoning is externally hosted and cannot be fully reproduced across model versions.** Hallucination risk and autonomous execution are bounded by the three-layer safety net described in Section 2.2. A residual reproducibility gap remains: the `claude-sonnet-4-6` API may change between model updates, and temperature-0 execution does not guarantee bit-identical outputs across versions. Future work will address this through (i) strict model-version pinning

in the API call, (ii) structured reasoning logs capturing the full LLM response per building, and (iii) standardized replay tests that re-run stored prompts and flag diagnosis drift relative to a reference set.

5. **Benchmark uncertainty is non-negligible for small CBECS strata.** Sparse categories produce wide EUI ranges, reducing the discriminative power of the median target for building-specific inference. Future work will test hierarchical benchmark fusion (regional + national priors), percentile-robust objective formulations, and sensitivity-aware target selection to reduce dependence on a single sparse stratum.
6. **The current benchmark database covers commercial and institutional buildings only.** CBECS excludes residential buildings, which constitute the majority of urban stock by building count. Future work will integrate RECS (Residential Energy Consumption Survey; U.S. Energy Information Administration 51) as a parallel benchmark stream. The LLM classification agent would infer use class from street-level imagery, GIS attributes, and permit records, then route each building to the appropriate database (CBECS for commercial and institutional; RECS for residential). This enables full archetype coverage without manual building-type annotation.
7. **Hardware scalability is extrapolated beyond the observed workload.** Parallel execution on six buildings converges wall-clock time to the single-building maximum (≈ 92 min); projecting this linearly to an N -building portfolio on an N -core node has not been validated at larger scale. Memory bottlenecks, I/O contention, and EnergyPlus licence concurrency may break the linear assumption. Future work will benchmark the pipeline on portfolios of 20, 50, and 100 buildings to characterize these constraints empirically.

A longer-term goal is a living UBEM framework that ingests updated scan data, weather files, and occupancy signals, then continuously re-infers energy parameters as conditions evolve. Realizing this goal requires tighter automation of upstream geometry quality control and reproducible pipeline orchestration, which remain the two most manual components in the current workflow.

7. Conclusions

This paper introduces Geo2UBEM, an automated UBEM framework made for places where meter data is limited. It uses image-based urban representations that are easy to access. Most urban buildings do not have annual end-use meter records, but conventional energy-parameter calibration usually needs this information. Geo2UBEM solves this problem two main parts. In Phase 1, it converts urban geometry into EnergyPlus IDFs ready for simulation. In Phase 2, it uses a three-agent LLM-MAS to estimate energy parameters with CBECS benchmarks when meter data is not available.

Four principal contributions and findings emerge.

- Phase 1 is an automated scan-to-IDF process that includes façade-aware improvement.** Phase 1 introduces three components that close the gap from accessible urban imagery to simulation-ready EnergyPlus models: facade-aware WWR enrichment from street-level images (Panel D), a harmonized geometry-descriptor schema with provenance meta-data (Panel E), and automated IDF synthesis (Panel F). This correction is most important when the default 30% uniform glazing ratio misrepresents actual facades and therefore distorts envelope heat transfer.
- Phase 2 involves LLM-MAS three-agent energy parameter inference in the absence of meter data.** The Orchestrator–Diagnosis–Parameter-Tuning workflow is the core methodological contribution. Using only CBECS benchmarks and LLM-inferred parameter priors, the system achieved benchmark convergence for all 6/6 P1F buildings ($|\text{NMBE}| \leq \pm 5\%$; mean $|\Delta\text{NMBE}| = 1.2$ pp under cold-start re-initialization, confirming independence from initial parameter values). When metered data are available, replacing the CBECS target with measured EUI enables building-specific inference in the same agentic framework, without retraining or architectural changes.
- The observability structure of scan-based UBEM is characterized by** a key structural constraint identified through Sobol analysis: equipment density dominates annual EUI variance across all six buildings ($S_T = 0.797\text{--}0.994$), whereas envelope parameters recoverable from exterior geometry contribute negligibly ($S_T \leq 0.005$). This explains why benchmark-guided inference is needed in scan-based UBEM: the dominant drivers of energy use are operational and cannot be recovered from exterior geometry.
- Uncertainty-aware and scalable deployment.** A three-layer UQ protocol (Morris screening, Sobol decomposition, and Monte Carlo LHS) provides probabilistic EUI bounds over the full prior parameter space, with all six buildings passing the near-optimal range-assessment criterion (NMBE interval < 6 pp within ASHRAE $\pm 5\%$). Parallel execution across the six-building portfolio achieved a $4.1\times$ wall-clock speedup (377.0 min serial to 91.6 min parallel), supporting practical deployment at campus and district scale.

Several limitations remain. The cohort includes six buildings in one climate zone (CZ5A) and uses ideal-loads HVAC, so broader generalization requires additional validation. LLM behavior is also partially non-deterministic; future work should enforce model-version pinning, structured reasoning logs, and standardized replay tests.

Geo2UBEM demonstrates feasibility of fully automated urban building energy modeling without metered input. Using publicly available urban 3D geometry, the framework

generates nationally benchmarked, uncertainty-quantified EnergyPlus models. As its scalability depends on building count rather than meter availability, Geo2UBEM enables benchmark-driven energy modeling for extensive urban building stocks lacking building-level metered data.

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A. LLM Agent Prompt Templates

This appendix reproduces the verbatim prompt templates supplied to the `claude-sonnet-4-6` model at each stage of the LLM-MAS pipeline. Dynamic fields filled at runtime are shown in `{curly braces}`. The templates are provided in full to support reproducibility and to facilitate direct comparison with related prompt-engineering studies [27, 31]. Table 9: Prompt templates for LLM-based agents in the Geo2UBEM energy parameter inference loop.

Agent	Prompt
Diagnosis Agent	
System prompt	You are an expert building energy modeler specializing in physics-informed parameter estimation for EnergyPlus models. Your role is to estimate physically reasonable building parameters based on: - Building type and usage patterns (ASHRAE standards) - Climate zone characteristics - Engineering knowledge of typical energy use IMPORTANT: You are doing BLIND optimization. You do NOT have access to the building's actual meter data. Your target is the CBECS 2018 benchmark EUI for university laboratory buildings in Climate Zone 5. The final validation against real meter data will be done separately after optimization.
Diagnosis Agent	
User prompt	Analyze this building energy model against the ASHRAE/CBECS benchmark: **Building Information:** - Type: {building_type} - Name: {building_name} - Location: {location} - Climate Zone: {climate_zone} (Indiana – cold winters, humid summers) - Floor Area: {floor_area_m2} m² **BLIND OPTIMIZATION TARGET (CBECS 2018 Table C12 – EUI Distribution):** - P25 / Median / P75: {p25} / {median} / {p75} kWh/m²/year - Total Target (Median): {median_total_kwh} kWh/year - Acceptable Range [P25–P75]: [{p25_total} – {p75_total}] kWh/year **Model vs Benchmark Comparison:** {comparison_table_json} **Calibration Metrics (vs ASHRAE Benchmark):** - NMBE: {nmbe}% (target: +-5%) - CV(RMSE): {cvrmse}% (target: <15%) - Status: {PASS FAIL} **Available Parameters for Tuning:** equipment_density, lighting_density, occupancy_density, infiltration_ach **Your Task:** identify causes of deviation, recommend parameter adjustments, and output JSON with { "primary_issue", "parameters_to_adjust", "additional_considerations" }.
Tuning Agent	
Search-space prompt	You are optimizing EnergyPlus model parameters for calibration. **Building Context:** Type: {building_type}; Location: {location} (Climate Zone {climate_zone}) **Diagnosis Summary:** {diagnosis_summary_json} **CURRENT IDF Parameter Values:** {current_idf_params_json} Optuna must search NEAR current values, upward if model energy is too low and downward if too high. **Parameters to Optimize:** {parameters_to_tune_list} **Default Global Bounds:** {global_parameter_bounds_json} **Task:** define a narrow, physically reasonable range for each parameter and return JSON: { "parameter_name": { "min": <value>, "max": <value>, "rationale": "..."}, ... }. Be conservative to reduce simulation calls.
Tuning Agent	
Validation prompt	You are reviewing optimized EnergyPlus parameters. **Building:** {building_type} – {building_description} **Location:** {location} **Optimized Parameters:** {optimized_params_json} **Validation Checklist:** 1. Are all values within physically possible ranges? 2. Are they reasonable for a university lab building built around 1990–2000? 3. Are there suspicious combinations (e.g., very high U-value + very low equipment)? 4. Do equipment densities make sense for the specific building type? **Output JSON:** { "approved": true/false, "confidence": 0–1, "concerns": [...], "recommendations": [...], "corrected_params": {} }.

All three prompts use temperature = 0 to maximise determinism. The Orchestrator Agent contains no LLM calls; it is a pure Python state-machine that routes outputs between the two LLM agents, decides whether to iterate, and enforces the HITL confidence-gate (Section 2.2): any building whose Tuning Agent validation returns confidence < 0.7 or a non-empty concerns list is automatically routed to a structured human-review queue for post-run inspection.

A.1 Representative Agent Interaction — PMU Building

The following excerpt reproduces the actual Diagnosis Agent output and the resulting Optuna-tuned parameters for the Purdue Memorial Union (PMU), the single calibration iteration required to reach convergence, starting from the neutral-parameter baseline (Section 3.1), whose simulated EUI exceeded the CBECS space-weighted mixed-use composite benchmark of 606.4 kWh/m²/year by 15.1% (NMBE = +15.1%). This example illustrates the full reasoning chain from simulation metric to LLM-guided search-space definition and final convergence.

Diagnosis Agent response (verbatim, Iteration 1):

```
{
  "primary_issue": "Simulated EUI is ~15% above the CBECS median, suggesting internal loads (equipment, lighting) or infiltration are slightly over-estimated for a mixed-use university building that includes non-lab spaces.",
  "parameters_to_adjust": [
    {"name": "equipment_density", "suggested_action": "Decrease by 10-15% toward 15-25 W/m2 blended average for mixed-use union building", "confidence": "high", "priority_score": 9},
    {"name": "lighting_density", "suggested_action": "Decrease by 8-12% to ~9-11 W/m2 blended LPD reflecting mixed occupancy types", "confidence": "medium", "priority_score": 4},
    {"name": "infiltration_ach", "suggested_action": "Decrease to 0.15-0.25 ACH consistent with a renovated institutional masonry building", "confidence": "medium", "priority_score": 2}
  ],
  "additional_considerations": "The Purdue Memorial Union is primarily a student union, not a laboratory building – applying pure CBECS lab EUI benchmarks may inherently overstate loads; the mixed-use classification should weight common/dining/office spaces heavily. A 15% overage is modest and within P75, so aggressive reductions risk undershooting; target a combined 10-15% reduction across equipment and lighting densities."
}
```

Guided by this diagnosis, the Parameter Tuning Agent narrowed the search space toward lower equipment and lighting densities and reduced infiltration; Optuna TPE then searched this narrowed space directly against the CBECS benchmark (no metered data used). The best-trial parameters were: equipment_density = 29.64 W/m², lighting_density = 10.60 W/m², infiltration_ach = 0.225 (occupancy density was not flagged for adjustment and remained at the neutral baseline, 0.05 ppl/m²). The resulting NMBE was −0.04%, within the ±5% ASHRAE threshold; PMU converged after this single iteration (Table 5, Table 6).

CRedit authorship contribution statement

Wei Wu: Writing – review & editing, Writing original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ming Qu:** Writing – review & editing, Writing original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Supervision. **Zhe Wu:** Writing – review & editing, Visualization, Validation, Resources, Methodology, Investigation, Data curation. **Yongfei Li:** Visualization, Validation, Investigation.

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